Manual

Editing of Orders/Work Plans (MOC)

BDE-BAA 8.2

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Contents

1 Übersicht Bearbeiten Aufträge / Arbeitspläne

Purpose

The function package “editing orders/work plans” provides an extensive range of functions for

- creating or modifying orders with operations in the system
- creating and editing work plans in the system
- creating orders based on work plans

Implementation considerations

You use the function package if

- You have no interface to an ERP system from where orders are retrieved.
- You have an interface to an ERP system from where orders are retrieved and you
    - would like to modify the transferred orders/operations in MES,
    - want to create and manage your own orders in addition to the orders transferred from the ERP system.
- For activities that are performed over and over again, you want to use a work plan as a basis for creating new orders.

Features

- Order management
    - Function for creating production, overhead cost or rework orders
    - Assigning 1-n operations to an order
    - Assigning material components, production resources and documents to operations
    - Assigning long texts to orders and operations
    - Option to modify and correct order backlog data supplied by ERP/PPS systems
- Work plan management
    - Creating copy masters of work plans used to generate production, overhead cost and rework orders
    - Function used to generate production, overhead cost and rework orders from work plans
- Generating orders
    - Creating production, overhead cost or rework orders based on work plans
    - Creating production, overhead cost or rework orders by copying already existing orders

2 Order Object

Definition

An order in MES is generally a request to production to carry out a certain action accounting for a variety of work steps. A distinction is generally made between a production order and so-called overhead cost orders.

1. 1. The production order (production) defines:

- The material/ article to be produced
- The quantity to be produced (batch size)
- The earliest and latest start and finishing dates

2. 2. The overhead cost order (service) defines:

- A particular activity
- The calculation reference

The order in MES primarily consists of

- The order header information
- The operation data

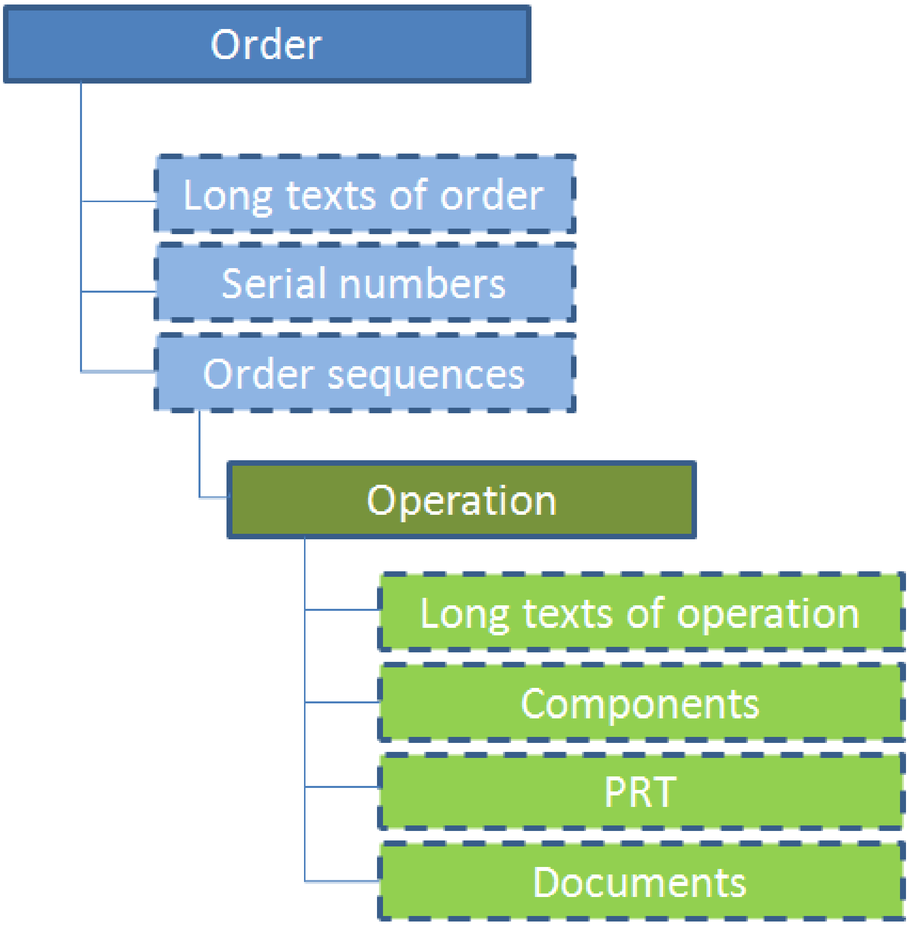
Usage

In MES the order (header) information is used to complete the shop floor papers and to manage the data, which is the same for any operation/process of a given order.

All activities that a person carries out on a machine/work station are order and/ or operation related. The posting of the order and operation answers the question what is being done and/or what activity is being carried out.

Structure

Every order is identified by a unique ID or order number. This is either provided and administered by an upstream system (generally ERP systems) or by the MES system itself. The object Order is structured as follows:



Integration

The order includes n operations that are to be carried out. The order thus produces a certain material or final product with a certain type of material.

3 Operation Object

Definition

The operation or procedure is one step within a work flow during which a manufactured quantity of an order's article is produced.

In addition to the operation number, the information listed below is needed to identify the operation:

- A written description of the work that needs to be performed
- The required workplace or the required group of identical workplaces
- In some cases, any other resources needed (e.g. tools, drawings, NC programs)
- Time needed to carry out the work (e.g. setup time, processing time)
- Target quantity (batch size)

If different machines or groups are required for production, this is what we refer to as multilevel production. Therefore, multilevel production includes several machine-related operations, which normally run one after the next. The number of operations needed for an order is not limited.

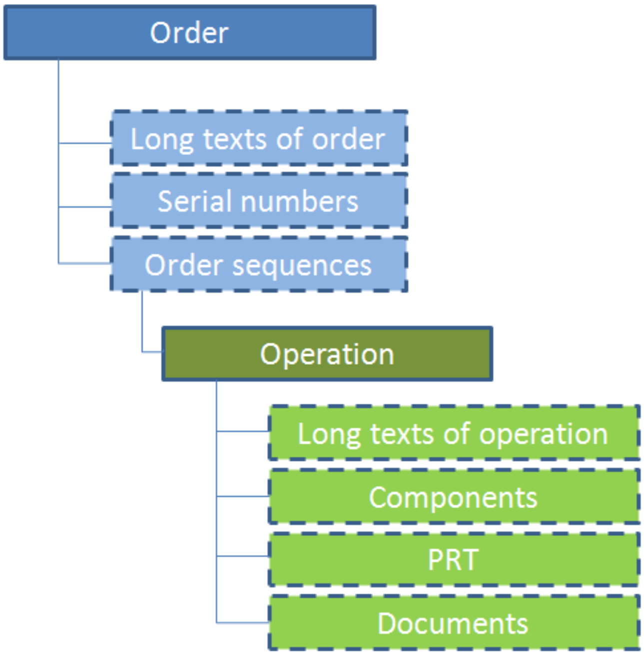
Terms used synonymously for the term operation are: procedure or order sequence/maintenance sequence (AFO). Oftentimes, the term order itself is also used synonymously.

Usage

All activities that a person carries out on a machine/work station are order and/or operation related. The posting of the order and operation answers the question what is being done and/or what activity is being carried out.

Structure

Each operation can be identified by the relevant combination of the unique order number and the sequence and operation number. This is either provided and administered by an upstream system (generally ERP system) or by the MES system itself. The object "operation" is subordinate to the object "order" and "sequence" and is structured as follows:



Please note: the object order sequence is only used if specifically requested.

Integration

The operation outputs a material with a specific material type. The operation also includes as additional information the bill of materials or rather the component list showing the materials that are needed or that are relevant in manufacturing the article. The same applies to the range of different production resources (e.g. tools) itemized in the production resources and tools list.

4 Edit Orders

Overview

HYDRA menu	Order Management → Order management → Edit orders
FEDRA menu	Detailed Scheduling→ Order management → Edit orders
Transaction code	edor
Function authorization	edor

Available user fields		
Where	**Object type/user field key**	**Source (type)**
Table	AUNR/SYSTEM	Order (MF-D)
How to configure user fields?	Which user field types are available?	

Purpose

This document provides a description of how orders can be created and edited on the client.

Integration

Typical applications that require orders to be edited include:

- Creating overhead costs orders
- Creating orders if no ERP system is available
- Correcting order inventory data

This document also describes the order structure, i.e. the fields relating to the order header.

Requirements

The following configurations must exist

- Order types

Selection criteria

The application provides the following selection criteria:

Order

This selection criterion refers to the order number. The application shows the selected order. You can also enter wildcards.

Order type

This selection criterion references the order type. All orders with the selected order type are displayed.

Article

This selection criterion references the article in the order header. The application shows all orders that include the selected article. You can also use wildcards.

Sales order

This selection criterion relates to the sales order defined in the order header. The application shows all orders assigned to the selected sales order. You can also use wildcards.

Project number

This selection criterion refers to the project number defined in the order header. The application shows all orders of the selected project number. You can also use wildcards.

Planned order

This selection criterion refers to the planned order stored in the order header. The application shows all orders of the selected planned order. You can also use wildcards.

Customer name

This selection criterion refers to the customer name (designation) defined in the order header. The application shows all orders of the selected customer name. You can also use wildcards.

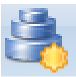
Checking the responsibility area

During the selection, the responsibility area defined for the order is checked.

Field descriptions

The separate fields in the order header are described here. The sequence described there may deviate from the sequence in the editing dialogs.

Toolbar



Generate order

Function authorization: or.generate

Starting the "generate order" dialog

Note: If you generate an order using this function, the work plan determination function is used. To generate an order from a specific work plan, please use the "generate order" function in the Work plan - Edit orders application.

Edit long texts of orders

Function authorization: edortx

Calling the application: Edit long texts of orders

Edit order sequences

Function authorization: edseq

Calling the application Edit order sequences

Edit operations

Function authorization: edop

Calling the application Edit operations

Order information

Function authorization: orin

Calling the application Order information

Order overview

Function authorization: orov

Calling the application Order overview

5 Edit Long Texts of Orders

Overview

HYDRA menu	Order management → Order management → Edit long texts of orders
FEDRA menu	Detailed Scheduling → Order management → Edit long texts of orders
Transaction code	edortx
Function authorization	edortx

Purpose

By applying the function “edit long texts of orders”, order-related additional texts can be displayed or edited. You use this function if:

- You would like long texts belonging to the order header to be visible and available in the administrative client while processing the order.
- You are using the MES Development Suite Label Designer component and the data you entered is to be printed on labels.

Keep in mind that for each order you use a maximum of one long text.

Integration

Long texts can also be transferred via the info interface (record type "AI"). Additional information about the interface can be found in the respective interface document.

Only long texts relating to the operation are displayed at the terminal.

Requirements

The corresponding order must already be defined.
Long texts included in the online data area may generally be edited, i.e. irrespective of the order status (added, modified or deleted).

Selection criteria

The application provides the following selection criteria:

Order

The long text for a specific order can be selected by entering the order number.

Field descriptions

The fields for long texts of orders are described here

Editing functions

To create a new operation or to edit one, you use the icons provided.

The long text entry function, which for the most part is equivalent to the functions of a text editor (highlighting of text passages; deleting or inserting of lines of text, as well as the merging of lines of text; copying with the key combination Ctrl+C, cutting with the key combination Ctrl+X, and pasting with the key combination Ctrl+V). Lines may have more than 80 characters when entered. When a document is saved, however, the system inserts a hard line break after the 80th character.

Toolbar

 **Edit orders**

Function authorization: edor

For the currently selected data record, this will call the application Edit orders.

6 Edit Order Sequences

Overview

HYDRA menu	Order management → Order management → Edit order sequences
FEDRA menu	Detailed Scheduling → Order management → Edit order sequences
Transaction code	edseq
Function authorization	edseq

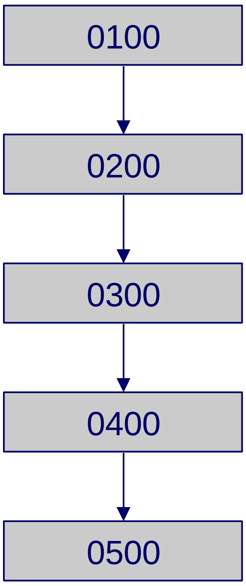
Purpose

If you want to group operations of an order, you can create sequences of operations in an order. In production, you use these sequences to specify the processing of operations. Within a sequence, the operations are processed one after the other according to the specified order. You can link several sequences. Using these links, you can integrate network-type structures.

You can also use parallel or alternative order sequences. The following sequence types are supported:

Standard sequence

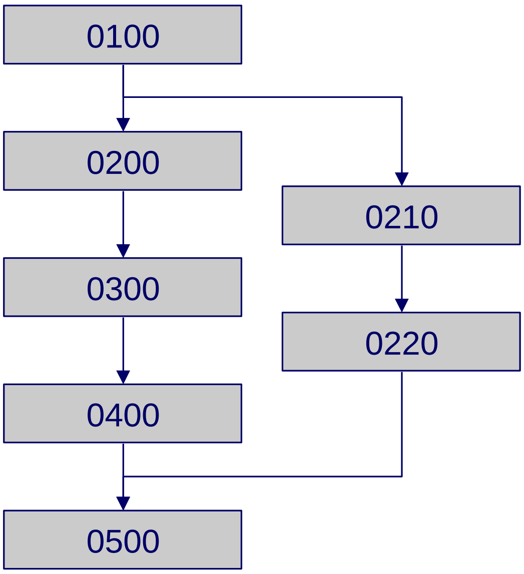
The standard sequence is available by default and describes the first sequence of the order.



If an order is sequentially processed, you only require the standard sequence. If you want to process specific operations in parallel or optionally to the standard sequence, you must create a new sequence including the respective operations. A standard sequence can then have parallel and alternative sequences. In this case, the standard sequence always has sequence number 0.

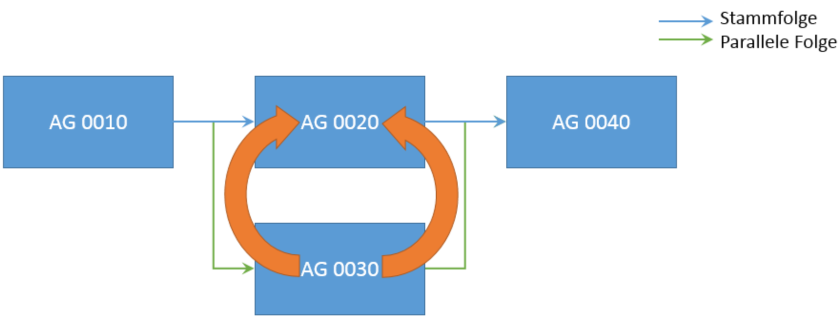
Parallel sequences

A parallel sequence runs in parallel to a partial sequence of the standard sequence. You use a parallel sequence, if specific processes must run at the same time. You use parallel sequences in process industry, for example.



In the example, the parallel sequence includes the operations 0210 and 0220. To position this parallel sequence, you must define an operation that identifies the start of the sequence (branch OP) and an operation that identifies the end of the sequence (return OP). You specify these operations in the application *Order sequences*.

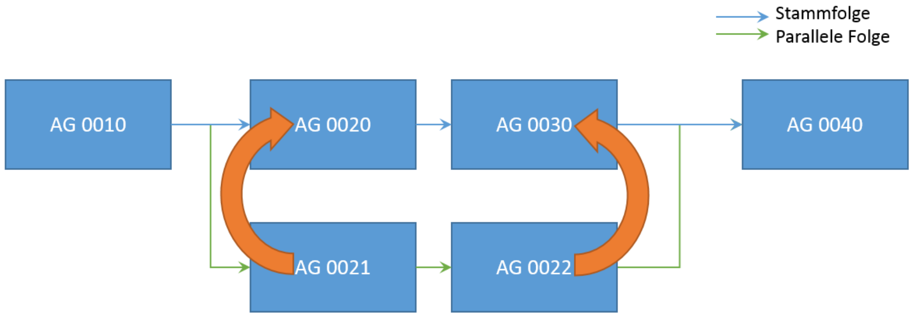
<u>Example 1:</u>



(blue = standard sequence; green = parallel sequence)

In the example, the parallel sequence only includes operation 0030. The respective branch OP for this parallel sequence is operation 0020, the return OP is also operation 0020.

Example 2:



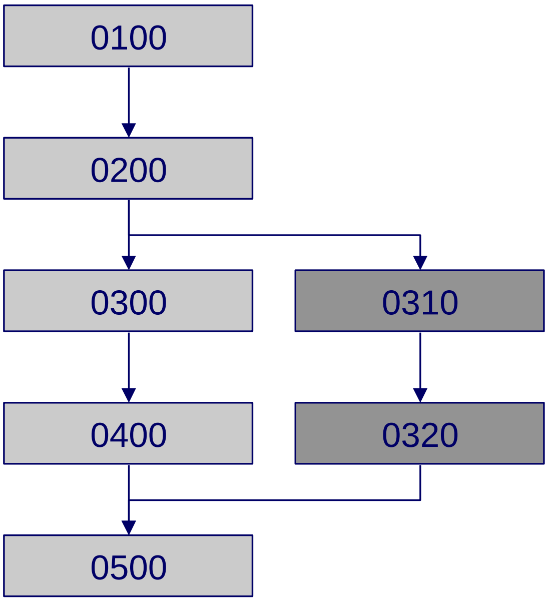
In the example, the parallel sequence includes the operation 0021 and the operation 0022. The respective branch OP for this parallel sequence is operation 0020, the return OP is operation 0030.

Alternative sequences

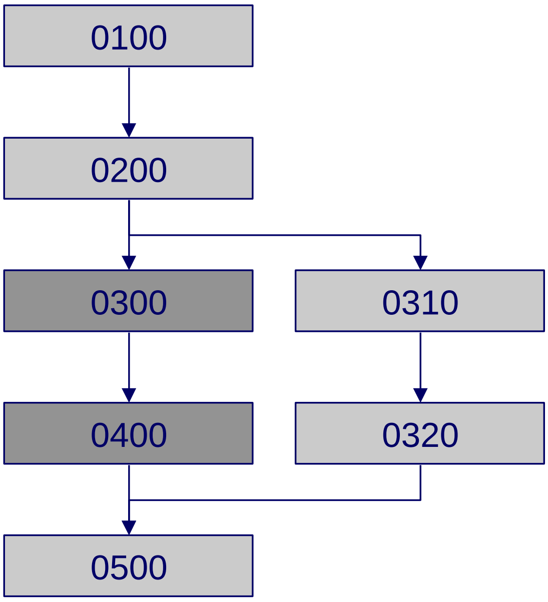
An alternative sequence describes one or more operations, which can be processed as an alternative to a partial sequence of the standard sequence. You use alternative sequences, if you have different production process for different batch sizes, for example.

If alternative sequences are available, only one of the alternative sequences is active and used for processing.

Order with an inactive alternative sequence



Order with an active alternative sequence

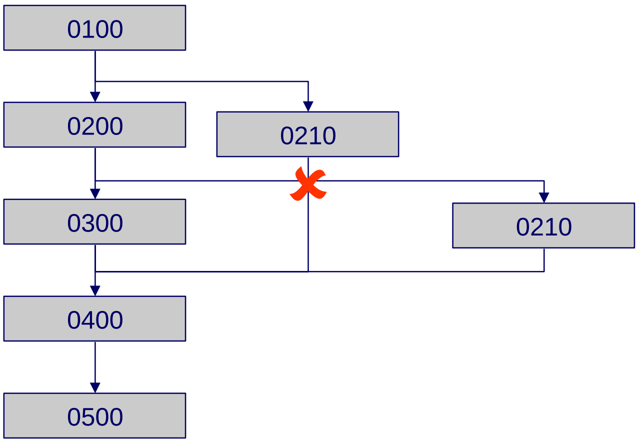


The system does not use inactive sequences. The inactive sequences are not used in scheduling or in detailed planning and you cannot make postings for inactive sequences.

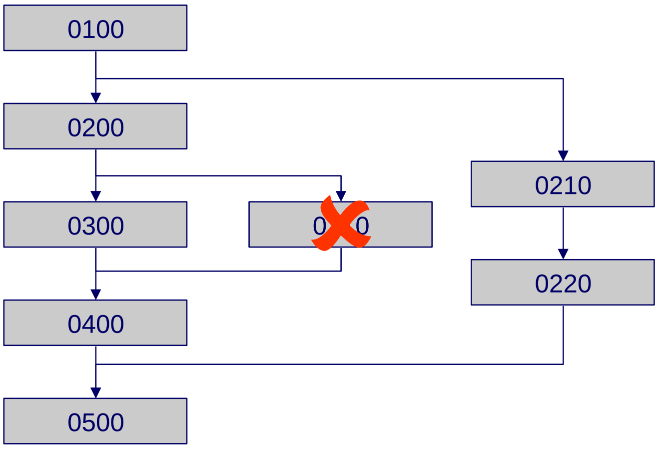
You can activate an alternative sequence on the client if specific conditions are fulfilled:

General

- Each operation is assigned to exactly one sequence.
- The system stores the resulting relationships between the operations and between the sequences in a table showing the internal relationships.
- Parallel sequences always run in parallel with the standard sequence.
- Alternative sequences are always an alternative to the standard sequence or to a part of the standard sequence.
- An overlapping of different operation sequences is not permitted.



- In case of a partial sequence of the standard sequence, which is specified using a branch OP and a return OP of a parallel or alternative sequence, another parallel or alternative sequence is not permitted that has a branch operation and/or return operation within the partial sequence.



Activating an alternative sequence

If you activate an alternative sequence, the respective part of the standard sequence is replaced by the alternative sequence. This partial sequence of the standard sequence is not run.

If the following conditions are fulfilled, you can NOT activate an alternative sequence:

- At least one operation of the partial sequence has already been started
- A second alternative sequence is already active in the partial sequence where this alternative sequence is included.

If you activate an alternative sequence, all other alternative sequences with the same branch and return OPs are deactivated.

If an alternative sequence is activated, the standard sequence is still identified as active.

Deactivating an alternative sequence

If you deactivate an alternative sequence, the respective part of the standard sequence is reactivated. This partial sequence is then run. If the following conditions are fulfilled, you can NOT deactivate an alternative sequence:

- At least one operation of the alternative sequence has already been started
- A second alternative sequence is already active in the partial sequence where this alternative sequence is included.

Deleting sequences

You can only delete a sequence that does not include any operations.

As a rule, you cannot delete a standard sequence.

Creating and deleting operations

If you create or delete an operation, the system automatically updates the order network for this order. The order network documents the relations between operations. The order network is used for the planning in the shop floor scheduling and for the processing/posting.

Copying an order

If an order is copied, the new order is available in its "initial state". This means that any existing alternative sequences are generally inactive, even if they were previously active in the order that was copied.

Operation status

Using the operation status, you cannot identify if the operation is part of an active alternative sequence or part of an inactive alternative sequence or part of an inactive partial sequence of the standard sequence (after activation of an alternative sequence). The operation status does not change during activation or deactivation.

Operations of active sequences have the initial status *prepared* when they are created. Operations of an inactive alternative sequence have the same status when they are created (*prepared*).

Sequencing list

The sequencing list does not display operations of an inactive alternative sequence or an inactive partial sequence of the standard sequence (after activation of an alternative sequence). You cannot log on these operations.

Merged operation

You cannot include operations in a merged operation that are part of an inactive alternative sequence or part of an inactive partial sequence of the standard sequence (after activation of an alternative sequence).

Specific features of parallel sequences

The planning algorithms in the shop floor scheduling also integrate parallel sequence structures.

If the option "Checking status of predecessor" is active and if you have combined several parallel sequences, you can only log on a succeeding operation when all sequences or more precisely the last operations of these sequences have been interrupted or finished. Any overlapping is identified.

To identify the send-ahead quantity of several parallel sequences, the system uses the smallest yield of the grouped sequences or of their last operations. The identified quantity is used as target quantity for the succeeding operations if the processing code of the respective operation includes a target quantity update.

Integration

To display operations of alternative sequences in the functions and evaluations on the client, note the following:

Operations / Operations logged on / Pool of orders

In the column *Control*, "Y" identifies an operation of an inactive sequence ("inactive operation").

If you do not want to display operations of inactive sequences in the order overview, then go to the selection panel, field *Control:* enable all options, but disable the option "Y".

Order overview

In tab *Progress*, the system displays operations of active and inactive sequences.

Order information

In the order information, the system displays operations of active and inactive sequences. If you want to identify the operations of inactive sequences, you must show the column *Control*. Use the column configurator of the operation table to show this column. In this column, the operations of inactive sequences have the identifier "Y".

Requirements

If you want to process sequences in the system, you require the respective license. You cannot use DOS based terminals (only applies when using HYDRA).

The following steps are required before use:

1. Configuration of the sequence number length in the system basic settings

WARNING

The sequence number length can only be configured during the initial system implementation process as long as no order backlog data is available in the system. A subsequent setting or change leads to inconsistent behavior.

2. Reactivation of the dynamic dialogs
   As a result, the input fields on the Windows terminal will expand by the defined sequence number length (only applies when using HYDRA).

Selection criteria

The application provides the following selection criteria:

Order

Enter the number of the order to display the sequences of this order.

Field descriptions

Order number

Order that includes the specified sequence. You can only create a sequence for an order if the order header is already available in the system.

Sequence

Identification of the sequence within an order.

Note: The standard sequence always has the sequence number 0.

If the "sequence" field is not shown in the editing dialog, the sequence number length in the basic settings is 0. Please contact MPDV.

Name

Description of the sequence.

Sequence category

S = Standard sequence
For each order, there is exactly one standard sequence; you cannot delete the standard sequence.

P = Parallel sequence of the standard sequence
For each order, several parallel sequences can be available

A = Alternative sequence of the standard sequence
For each order, several alternative sequences can be available

Note:
After having created a sequence, you cannot change the category of the sequence.

Active

The identifier "Active" is only relevant for alternative sequences:

J = Active

N = Not active

If you create a new alternative sequence, this sequence is set to *not active.*

In case of standard sequences and alternative sequences, this identifier is always set to *Active.*

Alignment

If several parallel sequences are available, the sequences usually have a different lead time. If you select a specific sequence, you can have a resulting time buffer. The alignment function controls whether these buffers are at the beginning or the end of the sequence. The following variants are possible:

F = Earliest due date
If you use the earliest date to align the sequence, the buffer will be at the end of the sequence.

S = Latest due date

If you use the latest date to align the sequence, the buffer will be at the beginning of the sequence.

N = Irrelevant; this is the case for standard sequences and alternative sequences.

If several parallel sequences are available for a standard sequence, the system checks the alignment for all parts of the standard sequence that have a parallel sequence.

Version

Revision number/version, for information purposes only.

Branch operation

Number of an operation included in the standard sequence:

- parallel sequence: the parallel sequence starts *before* the specified operation

- alternative sequence: the operation of the alternative sequence starts instead of the specified operation.

In case of parallel and alternative sequences, this is a mandatory field. In case of a standard sequence, this field must be empty.

If you manually create an alternative or parallel sequence, the branch operation of the standard sequence must already be available in the pool of orders. If you transfer a sequence via interface, you must transfer a valid operation number for this field (no validation check is performed).

Return operation

Number of an operation included in the standard sequence:

- parallel sequence: the parallel sequence returns to the standard sequence *after* this OP.

- alternative sequence: the alternative sequence replaces the standard sequence *up to* this OP.

In case of parallel and alternative sequences, this is a mandatory field. In case of a standard sequence, this field must be empty.

If you manually create an alternative or parallel sequence, the return operation of the standard sequence must already be available in the pool of orders. If you transfer a sequence via interface, you must transfer a valid operation number for this field (no validation check is performed).

Reference sequence

The reference sequence specifies the sequence of the order that is identified via the two operations, branch OP and return OP. The reference sequence is *always* the standard sequence (sequence number 0).

In case of parallel and alternative sequences, this is a mandatory field. The standard sequence must already exist.

In case of a standard sequence, this field must be empty.

Toolbar

Activate

Activate an alternative sequence

Deactivate

Deactivate an alternative sequence

Edit orders

Calls the application Edit orders.

7 Edit Operations

Overview

HYDRA menu	Order management → Order management → Edit operations
FEDRA menu	Detailed Scheduling → Order management → Edit operations
Transaction code	edop
Function authorization	edop

Purpose

The term process, work step or operation describes a workflow that is designed to perform a task in a work system. During this workflow one quantity unit of an order is produced.

You use this function to add new operations to an order or to edit data in existing operations.

Integration

Operations are planned in planning functions and optionally posted on shop floor terminals; their purpose is to facilitate status tracking and to record quantities and activities, which are usually uploaded to higher-level systems.

Requirements

The following requirements must be met when adding a new operation:

- The higher-level order must already have been created.
- If you use order sequences (project-specific; depending on the license), the sequence of the operation must already exist.
- The workplace/machine or group where you want to plan the operation has already been created in the system.

The authorization for the responsibility area is assigned to you and you are authorized to display the data.

Selection criteria

The application provides the following selection criteria:

Order

Enter the order number of the order that includes the operations you want to display. You can also use wildcards.

Operation

You can optionally enter the operation number of the operation that you want to display or edit. You can also use wildcards.

Sequence

If your system is set up to use sequences (depending on the license), you can enter the sequence number here. The system then selects the operations assigned to the sequence number entered. If your system is not set up for sequences, leave this field empty.

Show split OPs

If you use the function to split operations (requires license), you can use this option to define whether you want to display the split-master-operations only or also the included split operations.

Checking the responsibility area

During the selection, the responsibility area defined for the operation is checked.

Field descriptions

The fields of the operation are described here.

Only selected data is available in the table:

- Order
- Sequence
- Operation
- Split
- Processing code
- Locked
- Fixed
- Group
- Workplace

- Control

Editing functions

To create or edit operations, use the buttons provided.

If a responsibility area is stored for the order, the editing of data is only possible if the options to display, insert, modify and delete are enabled in the configuration of the responsibility areas or profiles.

Toolbar

Edit orders

Function authorization: edor.*

Calls the application Edit orders for the selected order.

Edit order sequences

Function authorization: edseq.*

Calls the application Edit order sequences for the selected order.

Edit long texts of operations

Function authorization: edoptx.*

Calls the application Edit long texts of operations.

Edit components

Function authorization: edopcomp.*

Calls the application Edit components.

Edit production resources and tools

Function authorization: edopres.*

Calls the application Edit production resources and tools.

Order information

Function authorization: orin

Calls the application Order information for the selected order.

Change operation status

Function authorization: op.statchg

Function to change the operation status.

Lock

Function authorization: op.lock

Use the button Lock operation to lock one or several selected operations.

Unlock

Function authorization: op.unlock

The button Unlock operation unlocks one or several selected operations.

Split operation

Function authorization: op.split

Calls the function to split the operation. For further information, refer to the relevant documentation.

Dissolve split OP

Function authorization: op.splitdissolve

Undoes the operation split. For further information, refer to the relevant documentation.

Adding an operation

Transferring order header data

The following data is transferred from the order header in the operation when a new operation is created:

- Order type
- Base quantity unit
- Article if the article number is not explicitly defined for the operation.
- Article designation if the article designation is not explicitly defined for the operation.
- Material type if it is not explicitly defined for the operation.
- Customer name
- Priority, if the priority control is set to *order-related* for the order type..

Any priority that may have been entered will be ignored!

Transferring default data

Default data is taken from a template or from the processing code, if one exists, and transferred to the operation when an operation is created. The data is transferred in the following order:

- Values are transferred from the template (if available).
  - If you add a new operation (manually or via interface), all values are transferred that can be edited in the template and that are not entered manually (explicitly) or transferred via interface .

- Values are transferred from the *processing code* (if one exists); doing so will overwrite any values set in the template. The following values are transferred from the processing code to the operation:
  - Underdelivery
  - Reaction to underdelivery
  - Overdelivery
  - Reaction to overdelivery
  - External processing
  - Recordable
  - Can be logged on several times
  - Can be split
  - Serial number obligation
  - Batch management requirement
  - Target quantity update*
  - Sequencing list* is no longer evaluated for display in the sequencing list; instead, the system directly accesses the separate configuration tables.

Note: Values marked with * are not displayed for the operation and therefore they also cannot be changed.

- Transfer of the values transmitted explicitly (either entered manually or transmitted via PPS interface); any values that were previously transferred from the template or the processing code will be ignored and overwritten.

Target quantity comparison

If the target quantity comparison is enabled for the preceding operation, then any target quantity that is entered is ignored and instead the target quantity of the preceding OP is used when you add an operation.

Identifying the transport time

To identify the transport time between two operations, you can store a transport matrix . This is part of a HYDRA customization. This transport time is then integrated during lead time scheduling.

When a new order or operation is created, the transport time is calculated using this matrix and then transferred to the operation. If you change the transport matrix later on, this will have no effect on already existing operations.

If a transport time is configured greater than zero in the ERP system and if operations are then transferred, these transport times are transferred to the database. Otherwise, the time is calculated using the transport matrix.

You need not use master data, you can also explicitly change the values for the operation. Note: any values changed explicitly are overwritten when you re-plan an operation to another machine group.

Setting the planned start data (used in the sequencing list on the terminal)

When a new data record is added, the system tries to identify planned start dates and to use them as default values for the sorting of the sequencing list. This process is based on the following logic:

- It is checked whether or not the planned start date and the planned start time are empty.
    - If yes:
        - The earliest start date (date) is used as planned start date.
        - The earliest start date (time) is used as planned start time.
- It is checked whether or not the planned end date and the planned end time are empty:
    - If yes:
        - The latest end date (date) is used as planned end date.
        - The latest end date (time) is used as planned end time.
- For the sorting of the sequencing list, the planned start date and the planned start time are entered in separate fields that cannot be changed.

The used date fields, the corresponding BAPI acronyms and database fields can be found in the document dealing with the technical background information on the sequencing list.

Editing an operation

If you edit an operation, the default data (if available) is taken from the template or from the processing code and transferred to the operation in the following order.

- If you change the planned group or if you change the workplace and the new workplace is included in a different group, then the system transfers the following values (if available) from the template:
  - Waiting time formula
  - Setup time formula
  - Processing time formula
  - Inspection time formula (only for information purposes).
  - Teardown time formula
  - Target cycle formula
  - Formula for the remaining run time
  - Formula for the second remaining run time
  - Max. synchronization time
  - Default value key
- If the planned group was changed, the system will also update the value plan_werk (internally managed in the order backlog) in which the company for the modified Group is identified and transferred.
- If the processing code was changed for the operation, the values of the processing code are transferred (see above).
- The final step involves the transfer of the values transmitted explicitly (either entered manually or transmitted via ERP interface); this will ignore/overwrite any values that were previously transferred from the template or the processing code.

If the group of an operation is changed, the transport time is recalculated. The transport time stored in the transport matrix is then used. Any transport time defined for the preceding OP will be ignored/discarded.

If you change the target quantity (P), the target quantity is only converted automatically to the other quantity units if the fields have been emptied manually beforehand.

Transferring order header data

If you change an operation, only the values below are transferred from the order header to the operations:

- Priority, if the priority control is set to *order-related* for the order type..

  Any priority that may have been entered will be ignored!

- Customer name

The base quantity unit is not changed, because it is very unlikely that a change of this kind would ever happen in reality. The material type is not modified, because in MPL it may vary from one OP to the next OP.

General checks run when an operation is saved

Checking the existence of workplace or group

If a workplace was entered, then the system checks whether the workplace exists. If yes, the workplace group of the workplace is transferred to the operation. In any other case, the saving process is interrupted with an error message.

If no workplace was defined, but instead only a group, then the system checks the validity of the group that was entered. This means: it checks if the group exists in the system. If no, the change is rejected and an error message is issued.

Checking priority management

If priority management was activated for the order type as part of the customizing process using the identifier with the same name ADE_AUFTRAGSARTEN.PRIO_STEUERUNG[2,2] and if the priority control was configured as order-relatedPRIO_STEUERUNG[1,1] = 'U' , then the system checks whether the defined priority is permitted when a new order is created manually or an order is changed. If the maximum number is violated, the action will be rejected.

If the order is transferred from the ERP interface and the maximum number is exceeded, the order is not be refused as a result of this validation check. In this case, however, the priority is automatically set to 1.

Ability to modify an operation

By default, the following operations cannot be modified:

- OPs that are currently logged on (status with control indicator L) and

- OPs that are automatically interrupted (status with control indicator F).

Checking the formula values transferred

As of b_anr.dll version 8.1.1.359.

If you add or change an operation, the system checks the values passed in the formula fields (if specified). The system checks if the values are available in the formula management. The validation check is performed for the following formula fields:

- Setup time formula (BAPI identification ANR.RUEZ:EXPR)
- Processing time formula (BAPI identification ANR.BEARBZ:EXPR)
- Inspection time formula (BAPI identification ANR.PZ:EXPR)
- Teardown time formula (BAPI identification ANR.ABRZ:EXPR)
- RRT1 formula (BAPI identification ANR.RLZ:EXPR)
- RRT2 formula (BAPI identification ANR.RLZ2:EXPR)
- Waiting time formula (BAPI identification ANR.WARTZ:EXPR)
- Target cycle formula (BAPI identification ANR.SZY:EXPR)

In addition to the formula fields, the following BAPI identifications are also checked:

- ANR.TE.EXPR
- ANR.TR.EXPR
- ANR.TEB.EXPR
- ANR.TRB.EXPR

If at least one of the values entered is not available in the formula management, the request is rejected and an error message is issued (return code 901).

You can deactivate the validation check using the following entry in the INI configuration:

Parameter name	**Value**
INI name	BDE
Section	ANR_BAPI
Key	CHECK_FORMULAS
Value	N → Nein / No (check is disabled)
Active	Yes
Comment	(optional)

Setting the planned start data (used in the sequencing list on the terminal)

When an existing data record is changed, the system tries to identify planned start dates, to update them and to use them as default values for the sorting of the sequencing list. The basic logic depends on the planning function configuration in the master record of the operation's workplace:

- Planning function "N" – no planning
    - It is checked whether or not the planned start date and the planned start time are empty.
        - If yes:
            - The earliest start date (date) is used as planned start date.
            - The earliest start date (time) is used as planned start time.
    - It is checked whether or not the planned end date and the planned end time are empty:
        - If yes:
            - The latest end date (date) is used as planned end date.
            - The latest end date (time) is used as planned end time.
    - For the sorting of the sequencing list, the planned start date and the planned start time are entered in separate fields that cannot be changed.
- Planning function "P" / "H" / "T" / "A":
    - If the planned start date was changed, the planned start date and the planned start time are entered in separate fields that cannot be changed.

If no workplace is defined for the operation, an identical processing is performed as with planning function "N".

The used date fields, the corresponding BAPI acronyms and database fields can be found in the document dealing with the technical background information on the sequencing list.

Deleting operations

When an operation is deleted, the following points must be considered:

- By default, you can only delete an operation if the operation is not logged on, i.e. the operation is not in status "running" and not in status "automatically interrupted".
- You cannot delete a split operation. To delete a split operation, you must dissolve the operation using the relevant split functionality.
- If a split master is deleted, the split OPs are also deleted.
- You cannot delete a merged operation. You must dissolve it.
- If you manually delete the last operation of an order on the client, the order header is not deleted. It must be deleted explicitly.

By default, deleting an operation means that an item is physically deleted from the database. The following data is deleted:

- Backlog of orders
- Order status
- Assigned material components
- Assigned production resources and tools
- Assigned long texts
- Resource allocation for this operation in the shop floor scheduling (HLS)

The log data (Tabelle ade_protokoll) is not automatically deleted if an operation is deleted. The log data is transferred to the long-term table or deleted from the database in the course of the cyclic archiving/deletion runs.

Deleting orders via delete action "D" if last OP is deleted

As of b_anr.dll version 8.1.1.358.

If you "delete" the order header via MLE interface (PPS=J), only physically deleted operations are integrated. For operations that were only deleted logically (using delete action=D), no processing is available.

Use the following entry in the INI configuration to integrate also operations that were only logically deleted (delete action=D):

Parameter name	**Value**
INI name	BDE
Section	ANR_BAPI
Key	ORDER_BAPI_DELETE_WITH_DELETE_ACTION_D
Value	J → Ja / Yes:
Active	Yes
Comment	Delete Order when OP in status D

HYDRA: as of service pack >13

FEDRA: as of version 1.1

This processing is activated by default for new systems. You can deactivate this processing, if required. The processing is not automatically activated with subsequent updates.

8 Edit Long Texts of Operations

Overview

HYDRA menu	Order management → Order management → Edit long texts of operations
FEDRA menu	Detailed Scheduling → Order management → Edit long texts of operations
Transaction code	edtx
Function authorization	edoptx

Purpose

You can use the function *Edit long texts of operations* to display or edit operation-related additional. What should be considered in this regard is that only a maximum of one long text can be recorded/ assigned to an operation at any one time.

Operation-related long texts can be displayed on the terminal.

Long texts can also be transferred via the interface EIS-EZI (extension additional informations from ERP) (record type "AI"). Additional information about the interface can be found in the respective interface document.

Requirement

The corresponding operation must already be defined.
Long texts included in the online data area may generally be edited, irrespective of the operation status (added, modified or deleted).

Selection criteria

The application provides the following selection criteria:

MES order number

Entry of the combined order/ operation number. There is an option to use wild cards, for example in order to be able to display all an order's operation-related long texts. In this case, the order number must be entered, followed by *.

Field descriptions

MES order number

The operation's combined order/ operation number. This is a mandatory field.

Short Text

20-digit short text that is displayed in the table view. This is a mandatory field.

Long Text

The order's long text.

The long text entry function, which for the most part is equivalent to the functions of a text editor (highlighting of text passages; deleting or inserting of lines of text, as well as the merging of lines of text; copying with the key combination Ctrl+C, cutting with the key combination Ctrl+X, and pasting with the key combination Ctrl+V). Lines may have more than 80 characters when entered. When a document is saved, however, the system inserts a hard line break after the 80th character.

Toolbar

Edit operations

Calling up the application: Edit operations

9 Edit Notes

Overview

HYDRA menu	Order management → Order management → Edit notes
FEDRA menu	Detailed Scheduling → Order management → Edit notes
Transaction code	ednotes
Function authorization	edopnote

Purpose

You can use the function *Edit notes* to edit operation notes.

Requirements

The corresponding operation must have been created in the system.
You can generally edit notes included in the online data area, i.e. irrespective of the operation status (added, modified or deleted).

Selection criteria

The application provides the following selection criteria:

MES order number

Combined order/ operation number.

Please note that the components are assigned by specific operations. This is why the entire key must be entered. By entering the order number followed by *, the system will list all components for an entire order.

Field descriptions

MES order number

Combined order/ operation number.

Short Text

Short text of the note

Long Text

Long text of the note

Display on terminal

Specifies whether or not this operation note is shown on the terminal.

Toolbar

 **Edit operations**

Calls the application Edit operations.

10 Edit components

Overview

HYDRA menu	Order management → Order management → Edit components
FEDRA menu	Detailed Scheduling → Order management → Edit components
Transaction code	edcomp
Function authorization	edopcomp

Available user fields		
Where?	**Object type/user field key**	**Source (type)**
Tab *User fields*	MATLIST/depending on data record	Material component (MF-D)
Table	MATLIST/SYSTEM	Material component (MF-D)
How to configure user fields?	Which user field types are available?	

Purpose

Materials, which are required to produce an article, are assigned to an operation as (material) components. You can use the application *Edit components* to display or edit the material components of an operation.

Normally, the components are transferred to the system from the higher-level ERP system via interface, because the components are already defined in the ERP work plan.

Requirements

The relevant operation must have been created.

Selection criteria

The application provides the following selection criteria:

MES order number

Combined order/operation number.

Note: the components are assigned for a specific operation. For this reason, you must enter the entire key. To list all components of an order, enter the order number followed by *.

Field descriptions

The fields provided for a component are described here.

Toolbar

To create or edit a component, use the buttons *Insert* or *Edit*.

Note: when using the MPL, the BOM item must be **unique** for an operation.

Edit operations

Calls the application Edit operations.

Edit orders

Calls the application Edit orders.

Order information

Calls the application Order information.

11 Edit Production Resources and Tools

Overview

HYDRA menu	Order management → Order management → Edit production resources and tools
FEDRA menu	Detailed Scheduling → Order management → Edit production resources and tools
Transaction code	edres
Function authorization	edres

Purpose

Resources can be defined for operations in the list of production resources and tools.

Further information on how to define workforce requirements via production resources and tools can be found in the document entitled Definition_of_Workforce_Requirement.pdf

Requirement

The corresponding operation must already be defined.

Selection criteria

The application provides the following selection criteria:

MES order number

Combined order/ operation number.

Please note that the components are assigned by specific operations. This is why the entire key must be entered. By entering the order number followed by *, the system will list all components for an entire order.

Field descriptions

Order/ operation

Enter the order/ operation number for the operation that is to be assigned to the production resource or tool here.

Resource type

Resource type of the production resource or tool that is to be assigned to the operation. The resource type must be known in the system. Predefined resource types must be chosen from the selection menu. Additional resource types can be defined when customizing HYDRA. For documents, the resource type to be entered here must be DOC.

Resource

Enter the resource number (material number) of the production resource or tool.

Designation

Here, you can enter a name for the production resource.

Comment 1/ C\comment 2

These are comment fields.

Required quantity/ unit

Resource quantity required to carry out the operation. When planning the operation in the shop floor scheduling, this number of resources is entered in terms of capacities. The quantity unit is only used as a comment.

Please note: In the shop floor scheduling, the quantity 0 is interpreted implicitly as quantity 1.

Path

When identifying a document as a production resource, the logical reference to the path is to be defined in the path configuration (menu: File > System administration > Paths). No path must be stored for DNC resources; it is determined based on the path stored for the resource type. The field should be left empty for all other production resources (only applies when using HYDRA).

File

When identifying a document as a production resource, the file name (including file extension) is to be entered here.

No file name must be stored for DNC resources; it is determined based on the file name defined for the resource. The field should be left empty for all other production resources(only applies when using HYDRA).

Modified by/ date/ time

Editor as well as the date and time the last change was made.

Please note with regard to documents: If a new document is assigned to an operation a file is only uploaded automatically, in case a file has been selected using the file selection dialog. The file selection dialog can be opened by the button next to the “file name” field.

In this case, the path of the file that is loaded onto the server is displayed below the input field for the file name. The upload is performed automatically while saving.

No file can be uploaded if the file name is entered manually.

The corresponding data record is created anyway even if an error occurs during the upload.

Toolbar

Edit operations

Calls the application Edit operations.

Edit orders

Calls the application Edit orders.

Order information

Calls the application Order information.

12 Edit Order Network

Summary

HYDRA Menu	Order Management → Order management → Edit order network
FEDRA menu	Advanced Process Modeling → Edit → Edit order network
Transaction code	ednet
Function authorization	ednet

Usage

You use this application to create dependencies for orders beyond the existing operation sequence. These dependencies are referred to as relationships.

Keep in mind that only the end-start relationships can be created. These are relevant for both planning and for data entry. Enter the MES order number (combined order/ OP number) during data entry.

Requirement

The linked orders, including all of their operations, must exist in the system.

Selection criteria

The application provides the following selection criteria:

Order

The relationships are displayed for the selected order number.

OP

The relationships are displayed for the selected operation.

Predecessor/ successor/ predecessor and successor

Only the relationships relating to the selection are displayed.

Toolbar

This application only allows relationships to be created or deleted.

Any relationships created by the system automatically (origin = "S") may not be deleted by the user.

Field descriptions

Predecessor

Order number of the preceding operation

Preceding OP

Operation number of the preceding operation

Successor

Order number of the succeeding operation

Succeeding OP

Operation number of the succeeding operation

Relationship

Only the end-start relationships ("ES”) can be created in the setup process.

Origin

Relationships created manually or explicitly via the interface are created using "E" = externally created.

The relationships created by the system are marked with "S".

Active

In principle, relationships are always active. Relationships created due to alternative sequences are the exception. Relationships of inactive alternative sequences are marked as inactive.

Relevance

The system differentiates between relationships for planning and relationships for data entry.

P Relationship is only relevant for planning.

V Relationship is only relevant for data entry.

X Relationship is neither relevant for planning nor for data entry.

<empty> Relationship is relevant for planning and for data entry.

Explicitly set relationships can only be created with relevance =<empty>.

13 Work Plan - Edit Orders

Overview

HYDRA menu	Order management → Routing management → Work plan - edit orders
FEDRA menu	Detailed Scheduling → Order management → Work plan - edit orders
Transaction code	edwor
Function authorization	edwor

Available user fields		
Where	**Object type/user field key**	**Source (type)**
Table	AUNR/SYSTEM	Work plan - order (MF-D)
How to configure user fields?	Which user field types are available?	

The "Work plan - edit orders" application provides the user with a comfortable option to create or change work plans and to generate orders from work plans. A work plan is a kind of "empty envelope" for orders and is used to generate real orders.

Selection criteria

The application provides the following selection criteria:

Work plan

You can select a specific work plan if you directly enter the work plan number.

Order type

Use the combo boxes to select work plans of specific order types. You can check several options.

Article

Use the article field to search for work plans for a specific article. You can also use wildcards.

Sales order, project number, planned order

Using these fields, you can search by inventory data of the order header. You can also use wildcards.

Customer name

If work plans have been created for separate customers, you can search by the "customer designation". You can also use wildcards.

Field descriptions

Order header fields are described here

Notes

- Only selected data is available in the table:
  - Work plan
  - Order type
  - Article
  - Article designation
  - Target quantity (B)
  - Target scrap (B)
  - Unit (B)
  - Customer name
  - Sales order
  - Planned order
  - Project number
- The below-mentioned values cannot be edited in the work plan order:
  - Basic start date
  - Basic end date
  - Scheduled start time
  - Scheduled end time

Editing functions

Please use the available buttons to create or edit work plan orders.

If a responsibility area is stored for the order, the editing of data is only possible if the options to display, insert, modify and delete are enabled in the configuration of the responsibility areas or profiles.

Toolbar

Generate order

Function authorization: or.generate

You can generate an order from the currently selected work plan by calling this function. For further information on this, please refer to the section Generate order.

Edit long texts of orders

Function authorization: edwortx

Calls the application Work plan - edit long texts of orders.

Edit order sequences

Function authorization: edwseq

Calls the application Work plan - edit order sequences.

Edit operations

Function authorization: edwop

Calls the application Work plan - Edit operations.

Generate order

Please proceed as follows to generate an order from the work plan:

- Select the work plan, from which you want to generate an order, from the table.
- Open the function using the button . The "generate order" dialog opens.
- In the "order" field, enter the order number for the order that is generated. The field can be left empty if numbers are assigned automatically for the order type (customization).
- In the dialog, the input fields are populated with the work plan data. If required, change or add field values.
- Confirm the dialog by clicking .

An order is now generated from the work plan. By default, the application "Edit orders" opens with the new order that you have just generated.

Note: The *Edit orders* application is opened in a separate window with each order that is generated. It is therefore recommended to close the application before generating a new order.

If you want to suppress that the application *Edit orders* opens, you can configure this via INI configuration/INI data configuration. When the order is successfully generated, a popup informs that the "order xxxx has been successfully generated". Confirm by clicking OK.

Menu: System administration → System settings → INI configuration / INI data configuration

Name:	BDE
MOC user:	0
Comment:	Settings for Shop Floor Data Collection
Section:	EDWOR
Key:	GENERATE_ORDER
Value:	SUPPRESS_EDOR
Active:	☑
Comment:	Suppress automatic call of application "edit orders"

Notes

- The order cannot be generated if a responsibility area is specified for which the user is not authorized.
- The article number is transferred to the operations of the order.
- The order quantity and the unit are transferred to the order header as (basic) target quantity or target unit and to all operations as basic target quantity or target unit.
- The entered quantity is transferred 1:1 as primary quantity, if the primary quantity unit (primary unit of input) of the operations is identical to the unit that is specified as quantity unit above. If conversion factors are defined for the operation of the order or work plan to be copied, they are used to calculate the primary quantity. In case, no conversion factors are defined and the base quantity unit and primary quantity unit are different, the system tries to convert the base quantity into the primary quantity unit using an internal conversion table (system customization). This procedure generally also applies for the secondary quantity and the tertiary quantity.

14 Work Plan - Edit Order Long Texts

1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit long texts of orders
Transaction code	edwortx
Function authorization	edwortx

The "work plan - edit long texts of orders" function allows for additional texts to be collected for the order. There is only one long text at most for each order.

Please note

- Long texts can also be transferred using the HYDRA info interface (record type "AI"). For further information on the interface, please refer to the corresponding interface documentation.
- The terminal shows the long texts relating to operations only.

Selection criteria

The application provides the following selection criteria:

Work plan

The long text of a specific order of the work plan can be selected by entering the work plan number.

Field Descriptions

The fields of a long text pertaining to orders are described here.

Editing functions

Please use the available buttons to create or edit long texts of work plan orders. A copy function for order long texts is not planned.

Toolbar

Edit orders

Function authorization: edwor

Opens the Work plan - edit orders application for the currently selected data record.

15 Work Plan - Edit Order Sequences

¶ Summary

Menu	Order management --> Routing management --> Work plan - Edit order sequences
Transaction code	edwseq
Function authorization	edwseq

Operations are grouped in sequences to summarize them within an order. Production uses this information as an orientation tool to process each operation. Within the sequence, the operations are processed in sequence one at a time. By linking several sequences within the order, network-type structures can be illustrated

For further information on this, please refer to the document entitled "edit order sequences".

Selection criteria

The application provides the following selection criteria:

Work plan

The order sequences of a specific work plan may be selected by entering a work plan number.

Field descriptions

The fields of a sequence are described here.

Editing functions

Please use the available buttons to create or edit work plan sequences. A copy function for order sequences is not planned.

If the "sequence" field is not shown in the editing dialog, the sequence number length is 0 in the basic parameter settings. Please contact MPDV.

Toolbar

Edit orders

Function authorization: edwor

Opens the application work plan – edit orders for the currently selected data record.

16 Work Plan - Edit Operations

¶ 1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit operations
Transaction code	edwop
Function authorization	edwop

The "work plan - edit operations" application provides the user with a comfortable option to create or change operations for work plans.

Consequently, real orders may be generated from the work plans that are also often referred to as "empty envelope" for order orders.

Selection criteria

The application provides the following selection criteria:

Work plan

Operations of a specific work plan may be selected by entering the work plan number.

OP

The view may be limited to an individual operation of a work plan.

Sequence

This field is only relevant if sequences are used in HYDRA. In this case, the selection may be restricted to a specific sequence.

Field Descriptions

Operation fields are described here

Editing functions

Please use the available buttons to create or edit work plan operations.

The order (header) of the work plan must already exist in order to be able to create an operation.

Operations may only be copied within the work plan.

This table is to be used to document warnings/alerts (template).

Toolbar

Edit orders

Function authorization: edwor

Opens the application Work plan - edit orders for the currently selected data record.

Edit order sequences

Function authorization: edwseq

Opens the application Work plan - edit order sequences.

Edit operation long texts

Function authorization: edwoptx

Opens the application Work plan - edit operation long texts.

Edit components

Function authorization: edwcomp

Opens the application Work plan - edit components.

Edit production resources and tools

Function authorization: edwres

Opens the application Work plan - edit production resources and tools.

17 Work Plan - Edit Operation Long Texts

¶ 1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit long texts of OP
Transaction code	edwoptx
Function authorization	edwoptx

The "work plan - edit long texts of operation" function allows for additional texts to be collected for the operation. There is only one long text at most for each operation.

Please note

- Long texts can also be transferred using the HYDRA info interface (record type "AI"). For further information on the interface, please refer to the corresponding interface documentation.
- The terminal shows the long texts relating to operations.

Selection criteria

The application provides the following selection criteria:

MES work plan number

The long text of a specific operation of the work plan can be selected by entering the MES work plan number. The MES work plan number is the combined work plan/operation number.

Field Descriptions

The fields of a long text pertaining to orders are described here.

Editing functions

Please use the available buttons to create or edit operation long texts of the work plan. A copy function for operation long texts is not planned.

Toolbar

Edit operations

Function authorization: edwop

Opens the application Work plan - edit operations for the currently selected data record.

18 Work plan - edit components

Overview

HYDRA menu	Order management → Routing management → Work plan - edit components
FEDRA menu	Detailed Scheduling → Order management → Work plan - edit components
Transaction code	edwcomp
Function authorization	edwcomp

Available user fields		
Where	**Object type/user field key**	**Source (type)**
Tab *User fields*	MATLIST/depending on data record	Workplace - material component (MF-D)
Table	MATLIST/SYSTEM	Workplace - material component (MF-D)
How to configure user fields?	Which user field types are available?	

You can use the application *Work plan - edit components* to display and edit material components that are required to produce the article in the current manufacturing level (current operation).

The components are usually transferred via interface from the higher-level ERP system to the system, because the components are already defined in the ERP work plan.

Selection criteria

The application provides the following selection criteria:

MES work plan number

The components assigned to a work plan operation may be selected by entering the MES work plan number. The MES work plan number is the combined work plan/operation number.

Enter the whole MES work plan number if you want to show the components assigned to a specific operation.

If you want to show the components of all operations of a work plan, only enter the work plan number, followed by "*".

Field descriptions

The fields provided for a component are described here.

Editing functions

Please use the available buttons to create new or edit existing work plan components. A copy function for components is not planned.

Note: when using the MPL product group, the BOM item must be **unique** for an operation.

Toolbar

 **Edit operations**

Function authorization: edwop

Calls the application Work plan - Edit operations.

 **Edit orders**

Function authorization: edwor

Calls the application Work plan - edit orders.

19 Work Plan - Edit Production Resources & Tools

¶ 1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit production resources & tools
Transaction code	edwres
Function authorization	edwres

The "production resources & tools" application allows for the resources, which are required to produce the article in the current manufacturing level (current operation), to be displayed and edited.

Production resources and tools may be, for example, tools, documents, NC programs, etc.

Selection criteria

The application provides the following selection criteria:

MES work plan number

The production resources and tools assigned to a work plan operation may be selected by entering the MES work plan number. The MES work plan number is the combined work plan/operation number.

Enter the whole MES work plan number if you would like to view the production resources & tools assigned to a specific operation.

If you would like to view the production resources & tools of all operations of a work plan only enter the work plan number, followed by "*“.

Field Descriptions

The fields of a production tool and resource are described here.

Editing functions

Please use the available buttons to create or edit production resources & tools of work plans. A copy function for production resources & tools is not planned.

If the tool and resource management module (HYDRA-WRM) is in use, the first production resource and tool that is *not* of the resource type "DNC" or "MAT" is taken over into the "tool" field of the operation. In addition, the "tool" field is checked whether it already includes a value, when inserting a production resource and tool that is not of the "DNC" or "MAT" resource type. If this is not the case, this component is taken over. For this reason, it is recommended to insert the "main production resource & tool" at first in the list of production resources and tools.

Please note with regard to documents: If a new document is assigned to an operation a file is only uploaded automatically, in case a file has been selected using the file selection dialog. The file selection dialog can be opened by the button next to the “file name” field.

In this case, the path of the file that is loaded onto the server is displayed below the input field for the file name. The upload is performed automatically while saving.

No file can be uploaded if the file name is entered manually.

The corresponding data record is created anyway even if an error occurs during the upload.

Toolbar

Edit operations

Function authorization: edwop

Opens the application work plan - edit operations.

Edit orders

Function authorization: edwor

Opens the application work plan - edit orders.

- →→→→

20 Data Structure of Orders

This document describes each of the fields for an order header. The actual sequence of the editing dialogs and reports/overviews may deviate from the one illustrated here.

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

General **tab**

Order / work plan

The order number or rather the work plan number is an upper-level number, under which each of the operations is compiled.

Order type

Order types are issued to structure the orders in accordance with their use. Each order type includes various control information that is decisive when managing orders.

The glossary describes the order types that are available by default in the system. You can define additional order types when customizing the system.

Article/Item

Material number/ item number/ article numbers of the (final) article to be produced with this order. If no article is entered for the operation, the system transfers the article included in this field to the operations.

Article name

Name of the article. Any changes to the article name are transferred to all operations of the order (redundant information). You cannot edit the article name in relation to operations.

Drawing issue number

Drawing issue number of the article, also referred to as index (available as of BDE 8.2).

Customer name

Customer name

Sales order

Sales order number

Sales order item In addition to the sales order number, you can also enter the line item number of the sales order that this order refers to.

Priority

You can use the "priority" as a control tool for the order. The priority is a single digit, numeric value. The value increases in ascending order ("0" = lowest priority, "9" = highest priority).

This value specifies the color for the operation bar in the graphic planning board (Shop Floor Scheduling). The colored bar graphically indicates the priority of the operation. You can assign the colors to the priorities in the graphic planning board settings of the Shop Floor Scheduling product group.

During the customization process, you can determine based on the order type

- whether the priority of the order header is transferred unchanged to each of the operations

- whether priority management should be enabled.

Order index

The order index can be seen as an alternative to the priority. You can integrate the order index, for example, in sorting the graphic detailed planning (if configured accordingly).

The *order index* is numerical with a valid value range (-999.9 to +999.9).

Target quantity

Quantity specification for the production order in base quantity unit. The indicated target quantity may include a target scrap quantity that might have been entered.

Target scrap

Planned scrap quantity for the production order in base quantity unit. The indicated scrap quantity can be considered as part of the transferred target quantity.

Unit

Quantity unit of the order for the (final) article to be produced. The unit allows you to compare, for example, scrap from different operations. That is why, the unit is included as a base quantity unit in each operation (redundant information).

Material type

Material type of the (final) article to be produced. If the field does not include a material type, the MES inserts the value "SYSTEM" here.

Batch number

The batch number reserved for the order; is generally provided by the ERP system.

Dates tab

Basic start date

The basic start date of the order. In general, the ERP system specifies this date.

Basic end date

The basic end date of the order. In general, the ERP system specifies this date. This date is based on the required date/ delivery date set in the ERP system and, if necessary, also includes buffer times.

Scheduled start time

Scheduled start date; result of the lead-time scheduling as compared to infinite capacities.

If the scheduling is run outside of the system, the scheduled dates in the order header should be applied. If the scheduling is run in MES, these fields are overwritten.

Scheduled end time

Scheduled end date; result of the lead-time scheduling as compared to infinite capacities.

If the scheduling is run outside of MES, the scheduled dates in the order header should be applied. If the scheduling is run in MES, these fields are overwritten.

Scheduling type

The scheduling type describes whether the order is scheduled forward (V) or backward (R) during lead-time scheduling in MES. If scheduled forward, the order is scheduled based on the basic start date specified in the ERP system. If scheduled backward, the order is scheduled based on the specified basic end date.

If no scheduling type is set in the order, the scheduling type defined in *Basic Settings* is used.

Reduction strategy

If it turns out during scheduling that the lead time for a given order is longer than the allotted time available, then MES will attempt to take reduction measures to shorten the lead time accordingly. Reducible times are the waiting times and the transport times.

The document hls-bk.doc describes in the chapter *Reduction Strategies* how to configure reduction strategies. The configuration is performed as part of the customization process.

Assignment tab

Order group

If the field *order group* includes a value and the Priority control is enabled, the system checks the following when you attempt to create a new order:
- how many orders do exist for this order group and the specified priority in the system
- does this new order exceed the limit defined in the MOC application Order groups. If this new order exceeds the specified limit, the system will reject the order. If the priority control function is enabled, you must:
- consider the order group as a mandatory field in the order and
- define a preset value range in the MOC application *order groups.*

Note for SAP users

The term *order group* corresponds to the SAP production scheduler. To ensure a consistent data exchange between SAP and HYDRA, you should synchronize the possible SAP production schedulers with the MES order groups.

MRP controller

The MRP controller for the order. You can transfer the MRP controller from SAP to the MES for informational purposes. You can display the MRP controller in the MES. But the MES does not provide a predefined value range.

Project number

Project order number

Planned order

Planned order number, e.g. in serial production.

Cost object

Cost object number

Work plan

Work plan number of the work plan that served as the template for generating the production order.

Work plan version

Version number of the work plan that served as the template for generating the production order.

BOM version

Version of the bill of material assigned to the production order.

Production version

Production version on which the order is based. This field is currently only completed if planned orders are transferred via the HKMPP-REM interface.

Closed loop

ID of the closed loop / supply relationship for which a Kanban order has been generated.

Inspection order

Inspection order/ inspection batch number for the order

Sample type

Type of sample for the order

Calculation tab

The *calculation* index tab includes additional data fields where calculation-related values or information can be stored. These entries are for information purposes only.

Machine costs

Calculated value for the machine costs that are incurred in the production of this order.

Labor costs

Calculated value for the labor costs that are incurred in the production of this order.

Material costs

Calculated value for the material costs that are incurred in the production of this order.

Other costs

Calculated value for other costs that are incurred in the production of this order.

Material value

Calculated value of the produced final article for each base quantity unit.

Scrap value

Calculated scrap value for each base quantity unit.

User fields tab

User fields allow you to store further customer-specific information to the MES in addition to the fields that are available by default. The order information shows the order-related user fields. The order information dialog provides the *user fields* index tab for the order header. This tab shows the user field key, the defined user fields including the names and units of measure. The user fields tab includes eight sub-index tabs, which each have eight additional user fields. The so-called user field key determines which user fields are involved and which meaning they have.

User field key

Every user field key describes a combination of user fields. The management of the user field key (and therefore the purpose of the fields) varies from one object to the next.

User fields

The following user fields are available for the order header (object type AUNR) after customizing the system:

Field ID / index	Field data type	Number of fields
1 - 6	Date	6
7 - 22	Numeric, time, duration	16
23 -28	Decimal value	6
29 - 44	Text field, length 1	16
45 - 50	Text field, length 10	6
51 - 64	Text field, length 20	14
65 - 66	Text field, length 40	2

User field keys are defined in coordination with the customer during the customization process.

Administration tab

The administration index tab includes technical information on the data record. The dialogs "Insert" and "Copy" do not provide this index tab.

Created by

User who created the order.

Created on

Time and date when the order was created.

Modified by

User who most recently changed the order header.

Modified on

Time and date when this modification was made.

Transferred by

Here, you can enter the source from where the order was transferred.

Transferred on /Transfer time

If the ERP system transfers the order (PPS=J), the system automatically sets the transfer time and date to the time and date when the order was stored in MES.

Modified HYDRA

Specifies that the order was modified in MES. This identifier is automatically set to "J", if the order was changed in MES.

Modified PPS

Specifies that the production order was changed in the ERP system. This identifier is automatically set to "J", if the production order was changed in the ERP system (PPS=J). The identifier is not reset.

Deletion flag

Used for internal processing purposes. Cannot be modified.

Responsibility area

If a responsibility area is entered here, the user must have been authorized to view and edit orders and/or work plan orders.

21 Order Long Text Data Structure

Each of the fields for an order long text are described below. The actual sequence of the editing dialogs may deviate from the one illustrated here.

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

Order/ work plan

Order or work plan for which a long text is defined.

Short text

Short version of the long text, which is shown in the application list.

Long text

Actual long text, which is not shown in the application list.

The text entry function, which for the most part is equivalent to the functions of a text editor (highlighting of text passages; deleting or inserting of lines of text, as well as the merging of lines of text; copying with the key combination Ctrl+C, cutting with the key combination Ctrl+X, and pasting with the key combination Ctrl+V). Lines may have more than 80 characters when entered. When a document is saved, however, the system inserts a hard line break after the 80th character.

22 Data Structure of Order Sequences

Each of the fields for a sequence is described below. The actual sequence of the editing dialogs may deviate from the one illustrated here. Information about sequences can be found in the document edit sequences.

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

Order/ work plan

Order for which the sequence is defined. A sequence can only be set up for an order, if the order header already exists in MES.

Sequence

Identification of the sequence within an order.

Please note: The standard sequence always has the sequence number 0.

If the "sequence" field is not shown, the sequence number length is 0 in the basic parameter settings. Please contact MPDV.

Designation

Description of the sequence.

Sequence category

S = Standard sequence
There is only one standard sequence for every order; it cannot be deleted.

P = Parallel sequence to the standard sequence
There can be several parallel sequences for each order.

A = Alternative sequence to the standard sequence
There can be several alternative sequences for each order.

Please note:
The sequence category cannot be edited after a sequence has been set up!

Active

The qualification "Active" is only relevant for alternative sequences:

J = Active

N = Not active

If a new alternative sequence is set up, it is set as **not active**.

For standard sequences and alternative sequences, this qualification is always set to **active.**

Orientation

If there are several parallel sequences, the lead times generally vary in length. This creates time buffers in the sequences. The orientation function controls whether these buffers are at the beginning or the end of the sequences. The following options are available:

F = Earliest due date
If the sequence is set for the earliest date, the buffer will be at the end of the sequence.

S = Latest due date
If the sequence is set for the latest date, the buffer will be at the beginning of the sequence.

N = Not relevant; this is the case for standard sequences and alternative sequences.

If there are several parallel sequences for a given standard sequence, the orientation of the standard sequence is used for all segments of the standard sequence for which parallel sequences exist.

Version

Change number/version; for information purposes only.

Branch operation

Operation number of a standard sequence operation,

before which a parallel sequence should branch off, or

from which on an alternative sequence should be replaced.

This is a mandatory field for parallel and alternative sequences. For a standard sequence, this field must remain empty.

If manually setting up an alternative or parallel sequence, the branch operation of the standard sequence must already exist in the orders on hand. When a sequence is handed over via an interface, a valid order number also must be handed off (there is no plausibility check).

Return operation

Operation number of a standard sequence operation,

after which a parallel sequence should branch off, or

up to which an alternative sequence should be replaced.

This is a mandatory field for parallel and alternative sequences. For a standard sequence, this field must remain empty.

If manually setting up an alternative or parallel sequence, the branch-off operation of the standard sequence must already exist in the orders on hand. When a sequence is handed over via an interface, a valid operation number also must be handed off (there is no plausibility check).

Reference sequence

The reference sequence determines the sequence in the order that the reference operations (branch and return) refer to. This is **always** the standard sequence (sequence number 0).

This is a mandatory field for parallel and alternative sequences. The standard sequence must already exist.

For a standard sequence, this field must remain empty.

23 Data Structure of Operations

	→	

This document describes each of the fields for an operation. In this case, the index tabs specify how the fields are structured. The actual sequence may deviate from the one illustrated here.

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

General tab

Order / work plan

The order number or rather the work plan number is an upper-level number, under which each of the operations is compiled.

Sequence

The sequence number is the number of operation sequences in use.

OP

The operation number is the number listed below the order used to identify the operation.

Split

The split number.

OP name

Name of the operation; generally simply a short description of the activities that will be performed.

Article/Item

Part/item number of the article or material that is produced with the operation. If you do not enter the article, the system takes the value from the corresponding field of the order header.

Drawing issue number

Drawing issue number of the article, also referred to as index (available as of BDE 8.2).

Material type

Material type of the article that is to be produced in this particular production step. If you do not enter the material type, the system takes the value from the corresponding field of the order header.

Priority

You can use the "priority" as a control tool. The priority is a single digit, numeric value. The value increases in ascending order ("0" = lowest priority, "9" = highest priority).

Depending on the Order type, you can configure the priority to refer either to the operation or to the order. Choosing the latter will mean that the system will take the priority of the operation from the order header.

Planned on

This identifier indicates whether the operation is located in the group pool or if it is planned on a specific workplace / machine . In MES, you can plan operations either in the graphic detailed planning or via the order sequencing.

Entering or deleting the workplace later will NOT automatically change this identifier.

Planned workplace

If you set the identifier **planned on workplace**, this means that the operation is planned for the workplace entered here. If the input field is empty, the operation is not planned for any workplace.

Please note:
When you log on an operation, this field automatically includes the workplace where you logged in the operation. Doing so will overwrite any (in some cases a different) workplace for which the operation was planned up until then. As a result, the OP is implicitly re-planned.

Group

(Planned) station / machine group designated for producing the operation. It is meant as a planning criterion for group-oriented planning and in the graphic detailed planning.

If you log on an operation to a (different) workplace, its group will be updated, if necessary.

If, due to logging in the operation, there is a change to the group for which the operation was planned up until now, NONE of the values is taken from the template (this only happens if modifications are made manually via the editing function).

Fixed

This identifier specifies whether an operation is set as fixed during the planning process.

Before running automatic planning, the capacities (workplaces) are completely released with the exception of the fixed operations. Fixed operations that are still set in the past are moved to the right and set to "now" at the earliest plus a planning lead time. Any (fixed) operations planned for the future remain dispatched without changes.

Material

This field includes the first resource of the type "Material" (ID: "MAT") available in the component list. This is the "most important" input material.

You can use this field in planning, for example, when applying the Setup change list or in graphic detailed planning when planning equipment setup changes. It is of no significance to processing as part of the material and production logistics (MPL).

Color

You can enter the color of the main input material or the article planned for production here. This field is used in planning for example in the Setup change list or when planning equipment setup changes.

Tool

If you use the Tool and Resource Management (WRM) product group, this field includes the first resource available in the production resource / component list that is **not** of the resource type "DNC" or "MAT".

To this end, when a component is being entered that does **not** have a resource type "DNC" or "MAT", the system checks whether this field already includes a value. If the field does not include a value, this component is entered. For this reason, we recommend to first input the "main production resource" in the production resource list.

The graphic detailed planning integrates this field in order to identify production methods. However, the production resources stored in the operation are relevant when checking capacities.

By default, the field is of no relevance for processing in the Tool and Resource Management (WRM) product group.

DNC

If you use the production facility/resource management, this field includes the first resource available in the production resource / component list that is of the resource type "DNC" (ID: "DNC"). To this end, when a component with the resource type "DNC" is entered, the system checks whether this field already includes a value. If the field does not include a value, this component is entered.

In the system, this field is mainly used as a comment. By default, the field has no significance for DNC processing.

Upload number

The purpose of the confirmation/upload number is to identify an operation. This is a numeric value used for postings as an alternative to the combined order/OP number.

Examples

- Most of the time, a bar code is hard to read if the order / OP number is long (for example when using handheld barcode readers with a limited scanning range);
- If the space available on the work document is not large enough.

Please note: The length of the input field depends on the settings made for "Length of upload/confirmation number" in the basic parameter settings. If you did not specify a length there, the field is shown across the whole width of the application.

Leave this field blank for work plan operations.

Authorization

An authorization identifier that indicates whether a user is authorized to log on/off the operation. This involves crosschecking the identifier *OP postings* in the HR master.

Cost type

The cost type to be posted when executing this operation, for example in an overhead cost operation / order. At the moment, this field is only used as a comment.

Cost center

The cost center to be debited when executing this operation, for example in an overhead cost operation / order. At the moment, this field is only used as a comment.

Dates tab

The following dates are results calculated and executed during lead time scheduling. Lead time scheduling is triggered by certain events and runs asynchronously in the MES.

Scheduled start time

Scheduled start date of the operation as a result of the lead time scheduling compared to infinite capacities.

As a rule, a fixed operation is never changed. If, based on the scheduling situation, a date cannot be maintained, the operation is rescheduled, but it remains fixed.

Scheduled end time

Scheduled end date of the operation as a result of the lead time scheduling compared to infinite capacities.

Earliest start

Earliest start date (EST) of an operation as a result of forward scheduling during lead time scheduling as compared to infinite capacities or specified by PPS.

Earliest end

Earliest end date (EET) of an operation as a result of forward scheduling during lead time scheduling as compared to infinite capacities or specified by PPS.

Latest start

Latest start date (LST) of the operation as a result of backward scheduling during lead time scheduling as compared to infinite capacities.

Latest end

Latest end date (LET) of the operation as a result of backward scheduling during lead time scheduling as compared to infinite capacities.

Buffer time

The system determines the buffer time from the difference between the latest start date (LST) and the earliest start date (EST) for an operation

The sum total of the buffer times of all operations is stored in the order (header) in the field *OP buffer.*

Reducible time

If it turns out during scheduling that the lead time for a given order is longer than the allotted time available (basic end date exceeded), then the MES will attempt to take reduction measures to shorten the lead time accordingly. Reducible times are the wait times and the transport times.

This value indicates how many (more) hours can be reduced from the lead time of an order. This time results from the sum of the:
- difference from the current waiting time and the minimum waiting and the
- difference from the current transport time and the minimum transport time
Reducible time = (current waiting time - minimum waiting time) + (current transport time - minimum transport time). These differences are displayed here as totals.

The document entitled Reduction Strategies provides information on how to configure reduction strategies.

Planned start

Planned start date for the operation.

Logging in an operation that has not yet been planned will not result in the following:
- the time when this operation is started will not be interpreted as the planned start date and
- this date will not be entered here.

Planned end

Planned end date for the operation.

The planned dates (planned start/planned end) are set:

- by HYDRA Shop Floor Scheduling (HLS) upon saving
- by the Graphic Order Sequencing (GAV) upon saving
- by manual data maintenance (client)
- by the interface

The following logic applies for inserting and/or editing an operation (manual data maintenance or by interface):

- Insert/copy operation
    - If the "planned start" field is empty, it will be assigned to the earliest start date by default.
    - If the "planned end" field is empty, it will be assigned to the latest end date by default.
    - In both cases, however, the operation is not planned (automatically), i.e. you can still replan the operation.

- Change operation
    - In this case, processing depends on the "planning function" option of the workplace:
        - N (no planning):
            - If the "planned start" field is empty, it will be assigned to the earliest start date by default.
            - If the "planned end" field is empty, it will be assigned to the latest end date by default.
        - In any other cases, the planned dates will not be set automatically through processing.

Quantities tab

Generally, you can enter quantities in four different quantity units for an operation. Enter the target quantity and the unit (as abbreviation) for each quantity unit. You can also enter a calculated "estimated scrap" quantity. These quantities can be specified
- by the PPS system or, if the target quantity update is activated,
- they may result from the quantity produced by the previous operation.

The letters in parentheses behind the field descriptions provide information about the particular quantity type.

(P) Primary quantity unit

(S) Secondary quantity unit

(T) Tertiary quantity unit

(B) Base quantity unit

Target quantity (P) / unit / target scrap quantity (P)

Use the primary quantity to enter data via the terminal (primary input quantity).

The indicated target quantity may include a target scrap quantity that might have been entered.

Send-ahead quantity

In order to illustrate any overlapping, you can define a minimum send-ahead quantity (in the primary quantity unit) for the (preceding) operation. You can start the following operation (overlapping) if at least the send-ahead quantity has been finished and posted. The system verifies the send-ahead quantity during data collection (when logging on operations). In addition, scheduling and detailed planning also integrate any overlapping.

You have to enable the relevant configuration in the order type in order to verify the minimum send-ahead quantity when logging on OPs. Configure the processing code accordingly to plan overlapping operations based on the minimum send-ahead quantity (or the lead time). You can enable this function in the processing code while customizing the system.

When checking the minimum send-ahead quantity, the system only takes into account the recorded yield that has been entered up until now (primary quantity unit).

No quantity conversion takes place. For this reason, make sure that adjacent operations have the same primary quantity unit.

Example:

Operation 0100	Target quantity 1000	Send-ahead quantity 50
Operation 0200	Target quantity 1000	

If you enabled the validation check for the send-ahead quantity, you can only log on operation 0200, once operation 0100 has produced a yield quantity (in primary quantity unit) of at least 50.

The system does not check the operation status of the preceding operation. You cannot log on the current operation, in case the preceding operation has already been finished, but the send-ahead quantity has not yet been reached.

Target quantity (S and T) / unit / target scrap quantity (S and T)

The secondary and tertiary quantity are considered optional, variable units (for example within the reel-based MES solution - RF).

The indicated target quantity may include a target scrap quantity that might have been entered.

Target quantity (B) / unit / target scrap quantity (B)

The base quantity unit is an objective description of the material used in an order. The base quantity unit allows you to compare, for example, scrap from different operations. The base quantity unit is in effect the quantity unit shown in the order header. Generally, conversions (for example when target quantities are updated) are made using the base quantity unit.

The indicated target quantity may include a target scrap quantity that might have been entered.

If you use a quantity type, make sure to set the correct quantity unit.
The system only converts the quantities based on the conversion factors in the index tab "quantities" if the relevant values you want to recalculate are "empty" (not "0"). Quantity fields that contain values will not be recalculated.

Conversion factors

Use the conversion factors to convert the primary, secondary and tertiary quantities to the base quantity. Use these conversion factors, for example, when updating target quantities.

Use a numerator and denominator if you want to use decimal values (meaning a figure with decimal places) as the conversion factor.

Example

- Base quantity unit: Square meter M2
- Primary quantity unit: Piece PCE
- 1 piece = 2 square meters.

In this case
- define the numerator as the primary quantity 2 and
- define the denominator as the primary quantity 1

If no (valid) conversion factor exists, the system will attempt to convert the values using conversion formulas (this requires that formulas were defined during the customization process).

Overdelivery/ Underdelivery

The system checks all quantities you post for overdelivery. These quantities occur, when you report part quantities for an operation, when you interrupt or log off an operation. When you log off an operation, the system also runs a check for underdelivery.

For further information on overdelivery/underdelivery checking, see the document entitled MBL_PC_UnderOverDeliveryOverview.pdf.

Underdelivery (%)

Value shown as a percentage by which the quantity reported may deviate from the target quantity (primary quantity unit). The value is only assumed from the processing code if the value was not explicitly transmitted via the ERP interface.

Example:
Target quantity of the operation: 120 items
Underdelivery: 84%
The actual quantity must not fall below 101 items.

Overdelivery (%)

Value shown as a percentage by which the quantity reported may deviate from the target quantity (primary quantity unit). The value is only assumed from the processing code if the value was not explicitly transmitted via the ERP interface.

Example:
Target quantity of the operation: 120 items
Overdelivery: 168%
The actual quantity must not exceed 201 items.

Overdelivery reaction/ underdelivery reaction

If the limits specified in the fields *overdelivery* or *underdelivery* are exceeded, a warning or an error message may be issued in response. Possible values:

"empty" No reaction

W Warning

X Error.

If *error* is set as the reaction, you will not be able to override the validation check.

If *warning* is set as the reaction, you can override the validation check by entering a deviation reason.

Only Windows terminals allow you to enter a deviation reason. DOS terminals interpret the reaction "W" as an error.

Unit quantity

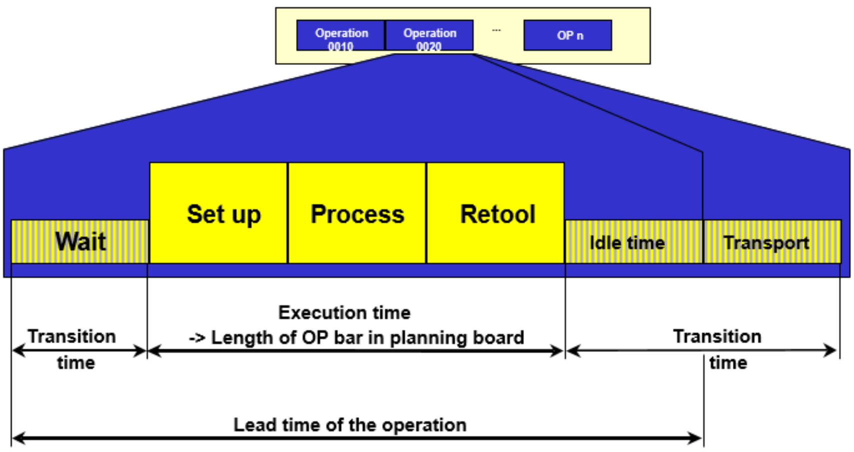
Quantity referring to operation specifications. You can customize the MES to use the ERP base quantity here. You can reference the ERP base quantity in formulas to calculate process times.

The unit of the unit quantity must be a primary quantity unit. The system does not perform an automatic conversion if the quantity units do not match.

As opposed to the base quantity in ERP, there is no other meaning or use in MES.

Durations / target times tabs

The illustration shown below provides an overview of the chronological structure of an operation in MES (in-house production).



Target setup time

The target setup time is the time required to prepare a workplace for the operation, for example the time needed to mount the necessary tools or to set the machine in compliance with the specifications ("setup time"). During this time, the workplace's capacity is shown as in use.

The ERP system transfers the target setup time or you can calculate the target setup time using a customized formula. The formula is based on default values. In this case, enter the formula in the field "setup time formula".

Additional setup time

The graphic detailed planning sets the additional setup time for the operation, if a setup change matrix is available and an additional setup time results from planning.
The additional setup time can also show a negative value.

Target processing time

The processing time is the time needed to process the material as part of an operation. During this time, the workplace's capacity is shown as in use. The processing time depends on the order quantity; it does include neither the setup time nor the retooling time.

The graphic detailed planning does not use the *processing time*. The graphic detailed planning calculates the processing time and/or remaining run time dynamically using the formula entered in the field "Formula RRT 1".

The ERP system transfers the target processing time or you can calculate the target processing time using a customized formula. The formula is based on default values. In this case, enter the formula in the field "processing time formula". Make sure to use the same basis to calculate the processing time and the remaining run time (field "Formula RRT1").

Planned retooling time

The planned retooling time (teardown time) is the time needed to reset the workplace back to its original state after the operation has been completed. This may require some tasks such as dismantling tools or performing some cleaning work. During this time, the workplace's capacity is shown as in use.

The ERP system transfers the planned retooling time or you can calculate the planned retooling time using a customized formula. The formula is based on default values. In this case, enter the formula in the field "Teardown time formula".

Planned delivery time

There is only one time component for external operations, i.e. the delivery time. The system synchronizes this time with the Gregorian calendar. The performance level has no relevance.

External processing

If this option is set, the operation is one that is performed externally. External operations are generally not planned, but only scheduled. In terms of lead time scheduling, for these kinds of operations the lead time only results from the delivery time.

If this option is not set, it will be considered an in-house operation. In this case, the following process times specify the capacity requirements (planning in HYDRA Shop Floor Scheduling):

- Planned setup time
- Planned processing time
- Planned retooling time

The following process times are used for scheduling (lead time scheduling) in-house operations:

- Target waiting time
- Target setup time
- Target processing time
- Target retooling time
- Target wait/idle time
- Target transport time.

Formula RRT1 / Formula RRT2

The value entered in the field *RRT 1* refers to a formula defined in the Management of formulas. The formula describes how to calculate the remaining run time (RRT) for an operation. The graphic detailed planning (HLS) uses this formula.

Unless otherwise specified or defined, enter the "RRT" value (remaining run time) in this field. In this case, calculate the remaining run time as follows (set by default):

(Target cycle / 1000) * (primary target quantity - the yield recorded up until now) / partitioning

You can enter another formula in the field *formula RRT 2.* You can use this formula to calculate any remaining run time that might deviate from *RRT 1.* The detail application *order progress* of the Order overview, for example, shows this formula.

Planned lead time

You can specify an overlapping of operations either using a send ahead quantity or lead time. The lead time describes the offset from the previous operation to its subsequent operation. A lead time can also be negative, if, for example, the subsequent operation begins with a setup before the previous operation.

Max. sync. time

If you enabled synchronization with the subsequent operation using the Processing code (customization), then planning makes sure that the maximum time span between this operation and the subsequent operation is the specified synchronization time.

The time is calculated in hours based on the shift calendar.

You can combine the synchronization function with an overlapping of operations.

Planned waiting time / waiting time formula / minimum waiting time

The waiting time is one available option to absorb interferences and delays for each operation. The waiting time describes the (calculated) length of time that needs to pass before an operation can commence (setup). The scheduling process also integrates the waiting time. You can enter the waiting time directly or you can calculate it using a formula.

You can reduce the waiting time during the scheduling process. For this purpose, you have to enter a reduction strategy for the order, which triggers a reduction in the wait time. You can reduce this time to the minimum waiting time (at most).

Target wait time

The target wait/idle time describes the length of time that needs to pass for processing-related reasons before a manufactured or processed material can undergo the next processing step. The scheduling process integrates the wait/idle time. You cannot reduce the wait/idle time.

Target transport time

The target transport time is the time necessary to transport material from one workplace to the next. The higher-level ERP system transfers the transport time or you can calculate the transport time in MES using a transport matrix.

Lead time scheduling takes into account the transport time. You can also reduce the transport time as part of lead time scheduling. You can define the transport matrix and reduction strategies when customizing the system.

Minimum transport time

If you use reduction strategies, you can reduce the transport time to this minimum amount of time during scheduling.

The following wage specifications are used to calculate an incentive wage.

Wage type

Wage type

Wage indicator

Piecework ID/premium (E/G/S/M/Z/...)

Target te

Premium default: te (per 1000 pieces).

"te" is the "time per unit" for each person. Use "te" to calculate the "order time", which is the specified processing time for each person used to calculate the incentive wage. By default, the MES shows this time in hours per 1000 pieces. The interface transfers this time in seconds per 1000 pieces.

If no incentive wage is used in MES, you can enter "0" here.

Target tr

Target tr is the person's default setup time (in hours).

If no incentive wage is used in MES, you can enter "0" here.

Target teb

The premium default "teb" is the available machine time per unit. You can use this time to calculate the "occupancy time" for the workplace/machine so that the incentive wage can be calculated. By default, the MES shows this time in hours per 1000 pieces. The interface transfers this time in seconds per 1000 pieces.

If no incentive wage is used in MES, you can enter "0" here.

Target trb

Target "trb" is the default setup time (in hours) of the workplace/machine.

If no incentive wage is used in MES, you can enter "0" here.

Processing tab

Processing code

A processing code is a compilation of options that are used to control the operations. Each operation references this kind of processing code, and as a result its performance is defined in relation to the issues listed below.

You can define processing codes at the time the system is customized. Unless defined otherwise, enter the Processing code SYSTEM.

Recordable

If you set this option, you can generally post the operation, provided other criteria are also met (e.g. operation not locked or operation can be logged in due to the status of the previous operation).

Can be logged on at the same time/parallel logon possible

This option specifies whether an operation may be logged on several times, i.e. at the same time.

You should enable this option for overhead cost operations and for operations that are logged on to group workplaces. However, you should not set this option for operations that are subject to batch management.

The planning functions graphic detailed planning, order sequencing, graphic order sequencing do not support operations that can be logged on simultaneously. These planning functions assume that an operation is planned for exactly one capacity. If you log on one operation to several workplaces at the same time, contrary to capacity planning, this is then in opposition to planning. In order to conduct parallel planning of operations on different capacities, MES provides the operation splitting function.

Batch management requirement

Set this option if the operation is subject to batch management. You have to use the material management in order to process operations that are subject to management in batches.

Serial number requirement

You should only set this option after consultation with MPDV.

Layout

The code entered here references a label that was created in MES Label Designer that needs to be printed for the operation.

Target cycle

The target cycle is an operation-related specification used for machine clocking. The target cycle does not depend on the number of produced parts. In MES, the target cycle is calculated and processed as a duration per 1000 machine cycles.

If cycle time monitoring is active, this value is assumed as the default setting for finishing the operation. This value is the default value for the MDE machine monitoring function (cycle monitoring).

Partitioning

The partitioning (cavity) defines how many parts are produced during a machine cycle.

The partitioning is determined for each operation separately and is transferred via the ERP interface to MES. The partitioning is transmitted to the terminal at the time the operation is logged on and applies to the machine to which the operation is logged on.

Pulse factor

Automatic quantity collection at the terminal includes the pulse factor that is stored for an operation. Consequently, the pulse factor and the partitioning represent a conversion factor for the automatic collection of quantities: primary quantity = cycle * partitioning/pulse factor.

Split authorization

This option defines whether an operation may be split.

Max. number of splits

If an operation can be split, the system checks whether or not the split number entered by the user exceeds the value entered here. If this is the case, the split is rejected with an error.

M/O relation setup (machine/operator relation: setup)

Personnel requirements PEP (Personnel Scheduling) for setting up the operation.

Qualification (setup)

Unique qualification number from the qualification master data.

M/O relation production (machine/operator relation: production)

Number of employees required for production. By configuring the system during the customizing stage, you can define for each order type that only the number of persons specified here can log on. If several operations are logged on at the same time (in parallel), the maximum number of persons is equal to the total number of M/O relations for each separate operation.

In Personnel Scheduling (PEP), you can use this field to define the personnel requirements needed to produce the operation.

Alternatively to defining personnel requirements by way of the machine/operator ratio for the operation, you can also define these requirements in the production resources and tools. As opposed to the production resources and tools, you can only define one required qualification each for setup and production if the M/O relation is used.

The machine/operator relation (for setup and production) is only relevant for personnel scheduling if you have entered a qualification in the corresponding field.

Qualification (production)

Unique qualification number from the qualification master data.

Production method (variant)

Using production methods allows you to specify on which machine an operation can be planned, when the ERP system transfers order specifications. If you use the graphic detailed planning, you can apply the available production methods for detailed planning taking into account the specified times (target cycle, setup and retooling time) for each production method.

Here you can enter the key of the currently assigned production method.

Data identifier

Here, enter the data identifier if you use an Arburg Control System (ALS). This ID must be unique (key). If ALS is not used, leave this field empty.

CBM tab

This index tab is only relevant in connection with the reel-based solution using in the material management module.

General

Special indicators

Not used; this field must remain empty.

Number of reels

The planned total number of reels to be produced (parent roll and sub-rolls); no specific processing in MES.

Material properties

Input width

Reel input width in MM

Output width

Reel output width in MM

If multiple rolls are manufactured at the same time in one operation, this field indicates the sum total of the separate widths.

If branches are planned, the output width of the separate operations is set explicitly (no sum total is generated) for each operation ("parent" and "sub-roll" operations).

Seam width

Total seam width in mm

If several reels are produced at the same time in one operation, this field will contain the sum total of the separate seam widths.

If branches are planned, the seam width of the separate operations is set explicitly (no sum total is generated) for each operation ("parent" and "sub-roll" operations).

Surface per piece

Surface for a piece in MM2/PCE

Mass per unit area

Mass per unit area in G/MM2

Casing weight

This is where the casing weight for the sub-rolls is defined while cutting operations.
Unit: G

Cutting information

Cutting OP

Only relevant if the operation is a cutting operation

" "	No roll cutting
"T"	Roll cutting active (sub-roll numbering)
"M"	Cutting active (parent rolls are being produced again)

Branch OP

Identifies parent and subordinate operations for a planned branch.

"M“	Mother OP of a planned branch
"K“	Child (subordinate) OP of a planned branch

Mother OP

If a branch is planned, these fields allocate the branched off material to the relevant mother operation.

A mother operation must reference itself.

Please note: Enter the MES order ID (= MES order number = combined order / operation number).

Daughter rolls/cut

Number of planned daughter reels per cut.

If the cutting plan is not defined, 0 is entered here.

Daughter rolls/cut - total

For cutting operations (mother OP): number of planned daughter rolls per cut (encompassing all branched off material).

If the cutting plan is not defined, 0 is entered here.

User fields **tab**

User fields allow you to store further customer-specific information to the MES in addition to the fields that are available by default. The order information shows operation-related user fields. The order information dialog includes the user fields index tab for the operations. This tab shows the user field key, the defined user fields including name and unit of measure. The user fields tab includes eight sub-index tabs, which each have eight additional user fields. The so-called user field key determines which user fields are involved and which meaning they have.

Object type

The object type for operation-related user fields is AGNR (cannot be modified).

User field key

Every user field key describes a combination of user fields. The management of the user field key (and therefore the purpose of the fields) varies from one object to the next. User field keys are defined in coordination with the customer during the customizing process.

User fields

The following user fields are available for the operation after customizing the system:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6

Field data type	Number of fields
Text field, length 20	14
Text field, length 40	2

Each page shows a maximum of 8 fields.

Default values tab

You can define up to ten default values for each operation. You can use the default values, among other things, to calculate certain processing times using specified calculation rules. The default value key specifies the meaning of each separate default value .

Please note

We recommend **not** to change the default value key at the operation directly, because this might distort the meaning of the separate default values.

Default value keys are configured in coordination with the customer during the customizing process.

Administration tab

Created by/Created on

User who entered the operation and the time that the operation was entered. You cannot change these fields.

Modified by/Modified on

User who most recently modified the operation, and the time that this modification was made. You cannot change these fields.

Transferred by/Transfer time

Here, you can enter the source from where the operation was transferred. If the PPS system transfers the operation (PPS=J), the system automatically sets the transfer time and date to the time and date when the order was stored in MES. You cannot change these fields.

Modified HYDRA

Specifies that the operation was changed in MES. This identifier is automatically set at "J", if the OP was changed in MES. You cannot change the field.

Modified PPS

Specifies that the production order was changed in the ERP system. This identifier is automatically set at "J", if the production order was changed in the ERP system (PPS=J). You cannot change the field.

Deletion flag

This option is only displayed in the order information. You cannot change the option.

Responsibility area

If an area of responsibility is entered here, the user must have been authorized to view and edit operations and/or work plan operations.

The fields listed below are only displayed in the Order information. You cannot change these fields.

Locked / locked by / locked on

You cannot log in locked operations. The terminal does not show locked operations in the sequencing list, irrespective of how the status is configured.

Additionally, the user is shown who was the last to lock the operation and also the time and date when the operation was locked. These values remain even after the operation is unlocked. They are updated each time the operation is locked again.

Unlocked by/unlocked on

Shows the user who was the last to unlock the operation and the time and date when the operation was unlocked. These values remain even if the operation is locked again any time in the future. They are not updated until the operation has again been unlocked.

Locked for editing / locked for editing by / locked for editing on

Reserved; currently not used.

Reactivated by / Reactivated on

If an operation that has already ended is reactivated, the user is displayed here, who was the last to perform the reactivation and the time and date on which the reactivation took place.

24 Operation Long Text Data Structure

Each of the fields for an operation long text are described below. The actual sequence of the editing dialogs may deviate from the one illustrated here.

MES order number / MES work plan number

Combined order/ operation number and/ or work plan/ operation number of the operation for which a long text is defined.

Short text

Short version of the long text, which is shown in the application list.

Long text

Actual long text, which is not shown in the application list.

The text entry function, which for the most part is equivalent to the functions of a text editor (highlighting of text passages; deleting or inserting of lines of text, as well as the merging of lines of text; copying with the key combination Ctrl+C, cutting with the key combination Ctrl+X, and pasting with the key combination Ctrl+V). Lines may have more than 80 characters when entered. When a document is saved, however, the system inserts a hard line break after the 80th character.

25 Data Structure of Components

Production resources and tools are stored in the table mlst_hy with resource type "MAT". In the sections below, the different fields of a (material) component are described. The actual order of fields in the editing dialogs can deviate from the order used here. Not all fields that are listed here are displayed on the client or can be edited.

MES order number / MES work plan number

The component is assigned to the operation identified in this field. The field shows the combined order and operation number or work plan and operation number of the operation.

Material

Enter the material number of the material component.

Designation

You can enter the name of the material.

Comment 1 / Comment 2

These are comment fields.

BOM item

The BOM lists the different components of a product. The components are referred to as items. The number entered here specifies the position where the item is listed in the BOM. It is therefore possible that the BOM includes one material number several times. Using the correct BOM item, the data collection can still be uniquely assigned.

Note: when using the MPL, the BOM item must be **unique** for an operation. Each component must have a unique BOM item if several components are used in one operation. Two components must not have the same BOM item.

For the coil-based solution "RF", the BOM item specifies the position of the component in the layer structure.

BOM level

A component can also have several levels. If applicable and known, enter the BOM level here.

If you log on input batches via material management (MPL/TRT), you can only log on components of BOM level 0.

If you enter a BOM level > 1, the system automatically sets the component type (see second next field) to "I" (info component).

Material type

Material type of the material component. The material type controls the material-specific processing in the system.

Unless defined otherwise for a specific project, assign the material type SYSTEM here.

The material type must be available in the system (see configuration of Material types). If no material type has been entered, the system tries to identify the material component (requirement: the assignment of material to material type has been made). If the system cannot identify the material type, the system uses the material type "SYSTEM".

For info components (material type "I"), we recommend to define and assign a separate material type (e.g. INFO).

Component type

Possible values:

M Material component (default)
You usually enter "M" here. Other component types can be relevant for material management ("MPL").

I Info component
You can display info components in the bill of materials (BOM), but you need not log them on or off.

T Carrier material (coil-based production)
You can log on a maximum of one input batch as carrier material (T) or added material (Z) to the machine.

A Scrap/waste material (coil-based production)

Z Added material as alternative for the carrier (coil-based production)
You can log on a maximum of one input batch as carrier material (T) or added material (Z) to the machine.

Consumption type

The following collection options are available for material components. The definition of the different options and their use depend on the functions used.

N = None

This option defines that no consumption is collected for the material component. The material component is only displayed here.

The so-called info components (see above: component type) must be set to this option.

L = Retrograde/with batch reference (MPL/TRT, MPL-RF)

If this option is used, the material component is logged on and off as batch. The consumption calculation for this material component (retrograde, at input batch logoff) depends on the configuration of the material type the material component is assigned to.

D = Discrete

This option is relevant for *discrete consumption recording* (AIP-DVE). This type of material consumption recording requires a configuration that can be part of a customization at the customer's.

Use the option "L" if Material and Production Logistic (MPL) or Tracking & Tracing (TRT) is used.

For this component, the system calculates consumption using the quantity produced last and suggests this calculated quantity in a posting dialog. The consumption is posted for the component and a material movement is generated (goods issue from production). This material movement can be uploaded to the higher-level ERP system.

Replaceable

If this identifier (=J) is set, you can use a different material than the material planned for this component. You can only use a material of the same material type.

For the user on the shop floor client (MPL/TRT) a message is displayed. The user selects and logs on the relevant component.

Change necessary / Requirement to change output batch

With this option, an input batch change for a batch of this material forces an output batch change. The setting that is allowed for this option depends on the relevant component type (see above):

Component type	Allowed settings
M	☐ or ☑ possible
T, Z	Only ☑ allowed
I, A	Only ☐ allowed

Input quantity

Input quantity of the component per unit in primary quantity that is planned for the operation.

Unit

Quantity unit of input quantity

Input quantity in percent / Upper tolerance limit / Lower tolerance limit

Default: 0; should only be modified after consultation with MPDV.

Required quantity

Total quantity of the component that is planned for the operation.

The system calculates the required quantity when the display is called. The following formula is used for calculation :

Required quantity = input quantity of component x target quantity of OP in primary quantity unit

The required quantity is only displayed in the table and in the detail panel.

Resource type

Reserved. Not used.

UOM Spec. mass per unit area

Reserved. Not used.

Spec. mass per unit area

Reserved. Not used.

Planned article

Reserved. Not used.

Backflush

Reserved. Not used.

Required quantity (PPS)

The field *Required quantity* shows the total quantity required that is transferred from the ERP/PPS system. That is the quantity of the component that is required to produce the target quantity of the operation.

Consumption (total)

The column *Consumption (total)* shows the total consumption that has been posted for the relevant component. In this context, it does not matter whether the component is subject to batch management or discrete.

Upper-level component: BOM item / BOM level

Reserved. Not used.

Modified by / Modified on

Editor and date and time of the last change

User fields

You can define and use user fields for specific projects.

26 Production Resources & Tools Data Structure

Each of the fields for a production resource or tool are described below. The actual sequence of the editing dialogs may deviate from the one illustrated here.

MES order number/ MES work plan number

Combined order/operation number and/or work plan/operation number of the operation for which a production resource is defined.

Resource type

Resource type of the production resource or tool that is to be assigned to the operation. The resource type must be known in MES. Predefined resource types must be chosen from the selection menu. Additional resource types can be defined when customizing the system.

For documents, the resource type to be entered must be DOC.

Resource

Enter the resource number (material number) of the production resource.

Designation

Here, you can enter a name for the production resource or tool.

Comment 1/ Comment 2

These are comment fields.

Required quantity/ unit

Resource quantity required to carry out the operation. When planning the operation in graphic detailed scheduling, this number of resources is entered in terms of capacities.

The quantity unit is only used as a comment.

Path

When identifying a document as a production resource, the local reference to the path is to be defined in the Path Configuration.

No path must be stored for DNC resources; it is determined based on the path stored for the resource type.

The field should be left empty for all other production resources.

File

When identifying a document as a production resource, the local reference to the path is to be defined in the Path Configuration.

No file name must be stored for DNC resources; it is determined based on the path stored for the resource type.

The field should be left empty for all other production resources.

If a new document is assigned to an operation, it must be ensured that it exists at the stated location. No file is uploaded when a document is assigned!

Modified by/ Modified on

Editor as well as the date and time the last modification was made.